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Comprehensive Classifications of Mammalian Enteric Neurons Across Functional, Morphological, and Molecular Modalities

Table of Contents

Introduction and ENS Organization

The enteric nervous system (ENS) is the largest division of the peripheral nervous system, containing hundreds of millions of neurons organized into interconnected ganglia that form distinct plexuses throughout the gut wall. This "second brain" can function independently of the central nervous system to control gut motility, secretion, and blood flow through complex neural circuits. (LLM Memory)

The enteric nervous system represents a remarkable example of neural complexity within the gastrointestinal tract, containing an estimated 400-600 million neurons in humans - more than the spinal cord (Model-Generated). These neurons are organized into two main ganglionated plexuses: the myenteric (Auerbach's) plexus located between the longitudinal and circular muscle layers, and the submucosal (Meissner's) plexus situated in the submucosa (Model-Generated). A third, less prominent subserosal plexus exists on the serosal surface in some species and gut regions (Model-Generated). The ENS displays remarkable functional autonomy, capable of generating and coordinating complex motor patterns, secretory responses, and vascular regulation without input from the brain or spinal cord (Model-Generated). This independence stems from the presence of complete reflex circuits within the gut wall, including sensory neurons that detect mechanical and chemical stimuli, interneurons that process and integrate information, and motor neurons that control smooth muscle, secretory epithelium, and blood vessels (Model-Generated). The intricate organization of enteric neural networks varies significantly across different gut segments, from the esophagus to the rectum, reflecting specialized functional requirements for digestion, absorption, and waste elimination (Model-Generated).

Morphological Classifications: Dogiel Types

Enteric neurons are morphologically classified into three main Dogiel types based on cell body shape and process patterns, *with Type I neurons having short processes and Type II neurons having long processes, though modern studies typically recognize only the distinction between these two primary morphological classes. (9 sources)*

The morphological classification of enteric neurons was first established by Russian histologist Aleksandr S. Dogiel in 1896-1899, who described three distinct neuronal types based on their structural characteristics (Sayegh et al., 2013) (Magalhaes et al., 2021). **Dogiel Type I neurons** are characterized by flattened, slightly elongated cell bodies with an angled or star-shaped contour, featuring a single axon and 4-20 lamellar dendrites that extend only a short distance from the cell body (Magalhaes et al., 2021) (Dharshika et al., 2022). These neurons project to and communicate with adjacent ganglia within the plexus and the musculature (Dharshika et al., 2022).

Dogiel Type II neurons possess large, round or oval cell bodies with eccentric nuclei, and their surfaces are grooved by bundles of neural fibers (Magalhaes et al., 2021). The defining characteristic of Type II neurons is the presence of several long, smooth axonal processes that are emitted either directly from the cell body (multipolar configuration) or from a single initial process that branches into subsidiary axons (pseudounipolar configuration) (Magalhaes et al., 2021) (Natale et al., 2021). These neurons communicate with neurons throughout the gut wall, both within and between ganglia, and extend to the mucosa (Dharshika et al., 2022).

Dogiel Type III neurons, the least understood category, have large cell bodies with variable shapes and multiple dendrites (Sayegh et al., 2013) (Nikolla et al., 2025). However, modern morphological studies typically recognize only the first two categories, as this dichotomy effectively distinguishes intrinsic primary afferent neurons from all other enteric neuron types (Brehmer, 2021) (Wang et al., 2025) (Brehmer, 2021).

Contemporary morphological analyses using advanced tracing techniques have revealed additional complexity within these basic categories. Studies using viral transfection methods have identified three distinct morphological classes: monoaxonal neurons with small cell bodies (~20 μm diameter) representing 94.7% of neurons, multiaxonal neurons with large cell bodies (40-45 μm) comprising 2.5% of neurons, and an intermediate class with two axons and small cell bodies accounting for 2.8% of neurons (Caillaud et al., 2023). The first two classes correspond closely to Dogiel's original Type I and Type II classifications, respectively (Caillaud et al., 2023).

Importantly, these morphological distinctions correlate with functional properties: Dogiel Type I neurons typically correspond to motor neurons and interneurons with synaptic-type electrophysiological properties, while Dogiel Type II neurons are associated with intrinsic primary afferent neurons (IPANs) displaying AH-type electrophysiological characteristics (Dharshika et al., 2022) (Zhao et al., 2023). This morpho-functional relationship has been conserved across species, with similar patterns observed in guinea pigs, mice, and humans (Dharshika et al., 2022).

Evidence

(Sayegh et al., 2013)

Back to Basics: Regulation of the Gastrointestinal Functions

> *"four main classification methods have been utilized. These methods depend on the [1] morphology (different shapes) of enteric neurons, [2] the types of neurotransmitters or peptides that they may contain (also known as their chemical coding), [3] the electrical properties of the enteric neurons or electrophysiology, and [4] the function (e.g., sensory, motor, inhibitory and excitatory) of the enteric neurons). On the basis of morphology the Russian histologist Alexander Dogiel described three types of enteric neurons in 1896 and 1899. These types are: (A) Dogiel type I, (B) Dogiel type II and (C) Dogiel type III neurons. Dogiel type I neurons have small, irregular cell bodies with multiple short dendrites. Dogiel type II neurons have large, oval-shaped cell bodies with one or two long dendrites. Dogiel type III neurons have large cell bodies with different shapes and multiple dendrites. Electrophysiologically, there are two types of enteric neurons: (I) S-type (S stands for synaptic) which evokes a fast (e.g., millisecond) action potential and (II) AH-neurons (AH stands for after hyperpolarization) which evoke longer lasting action potential (e.g., seconds) compared with the S-type"*

> *"some enteric neurons contain the neurotransmitter acetylcholine. Such neurons are called cholinergic neurons and are generally stimulatory to gut activities. Other enteric neurons contain adrenaline (also known as epinephrine). These neurons are called*

adrenergic neurons and are generally inhibitory to gut activities"

"the enteric neurons can be classified on the basis of their function. The basic functions of enteric neurons are excitatory, inhibitory, sensory or motor. Excitatory neurons cause an increase in secretion if they innervate a gland or cause muscle contraction if they innervate a muscle. Inhibitory neurons cause a decrease in secretion or muscle relaxation. Sensory neurons detect luminal pH and pressure or temperature in the gut wall, while motor neurons innervate muscles and sphincters and cause contraction or relaxation. In general, Dogiel type I neurons and S-type enteric neurons are considered motor neurons, whereas Dogiel type II and AH-neurons are considered sensory neurons. The excitatory enteric neurons of the gut, which act in the orad (toward the mouth) direction, contain acetylcholine (Ach) and substance P (sub P), while the inhibitory enteric neurons of the gut, which act in the aborad (toward the anus) direction, contain Vasoactive Intestinal Peptide (VIP) and Nitric Oxide (NO)"

"myenteric and submucosal plexuses"

(Magalhaes et al., 2021)

Enteric nervous system and inflammatory bowel diseases: Correlated impacts and therapeutic approaches through the P2X7 receptor

"As proposed by Aleksandr S. Dogiel in 1899, the morphological classification of enteric neurons can be based on their conformation and dendritic distribution. Dogiel described type I cells as flattened, slightly elongated, with an angled or star-shaped contour, and, as remarkable characteristics, as having only one axon and four to 20 Lamellar dendrites that frequently extend at a short distance from the cell body [31]"

"Type II neurons have large round or oval cell bodies and eccentric nuclei (Pompolo et al., 1988), and the surface is grooved by bundles of neural fibers (Hendriks et al., 1990). The main characteristic of type II neurons is the presence of several axonal processes that are emitted either directly from the cell body (multipolar neuron) or from a single initial process that branches into short subsidiary axons (pseudounipolar neurons) (Furness, 2012)(Song et al., 1994)"

"Additionally, enteric neurons can also be identified as intrinsic primary afferent neurons (IPANs), interneurons, and motor neurons (Furness, 2012), classified into at least 18 subtypes and using more than 30 neurotransmitters in their synapses [28]30]. Of these neurotransmitters, acetylcholine (ACh) and nitric oxide (NO) stand out as the most abundant (Brookes, 2001), as well as adenosine-5'-triphosphate (ATP) (Hansen, 2003), vasoactive intestinal polypeptide (VIP), and substance P (SP) (Gallego et al., 2016)"

"IPANs (classified as Dogiel type II) are recognized for responding to chemical stimuli, mucosal deformation and GI muscle tension, translating these signals into a neural impulse that will trigger a local motor reflex (Costa et al., 2000). Altogether, IPANs represent approximately 14% and 30% of the neurons of the submucosal and myenteric plexuses, respectively"

"The enteric neural circuit is arranged in two plexuses: the submucosal plexus, which in large mammals is present in two individualized levels (outer/inner) and is located in the outer connective tissue layer and the inner mucosal layer, and the myenteric plexus, located between the longitudinal and circular muscle layers (Bayliss et al., 1899)(Furness, 2012)5,28]"

(Dharshika et al., 2022)

Enteric Neuromics: How High-Throughput "Omics" Deepens Our Understanding of Enteric Nervous System Genetic Architecture

"Type I neurons have flat, elongate, and irregular cell bodies with a single axon and numerous short dendrites, whereas type II neurons have smoother and larger cell bodies with multiple long axons. Type I neurons project and communicate with adjacent ganglia in the plexus and musculature, whereas type II neurons communicate with neurons throughout the gut wall, within and between ganglia, and the mucosa"

"are also classified by electrophysiological properties, which have been mainly characterized in guinea pigs and mice. Two main types of enteric neurons are categorized as having either synaptic-or AH (after hyperpolarization)-type electrophysiological properties that differ on the basis of action potential speed and magnitude, the length of AH potentials, and tetrodotoxin sensitivity. Synaptic-type neurons typically display Dogiel type I morphology and include interneurons and motor neurons, whereas AH-type neurons typically display Dogiel type II morphology and are considered sensory neurons"

"Defining the neurochemical coding of enteric neurons was a significant advancement in identifying neuronal subtypes and understanding how enteric neurons communicate with one another and target cells. Enteric neurochemical coding has been defined by multiple approaches including immunohistochemistry in combination with retrograde tracing, electrophysiology, and pharmacology. Integrating these biomolecular data with morphologic and electrophysiological properties is the basis for current definitions of enteric neuron subtypes, which include motor neurons, interneurons, and sensory neurons. Although these definitions are based largely on studies in guinea pigs, many of the core features of enteric neuron subtypes are conserved between mice and humans. Excitatory and inhibitory motor neurons reside in the myenteric plexus and innervate the circular and longitudinal muscle of the intestine."

(Natale et al., 2021)

The Baseline Structure of the Enteric Nervous System and Its Role in Parkinson's Disease

"Intrinsic neurons of the ENS can be classified according to different criteria and approaches: morphological features (neuron size and distribution, and axonal projections, by means of light and electron microscopy), electrophysiological properties, and biochemical markers (neurotransmitter content and receptors, by means of histochemistry, immunohistochemistry, and transcript analysis)"

"Several types of intrinsic neurons that belong to the ENS have been described (Figure 3). These include intrinsic primary afferent neurons (IPANs) possessing chemo-and mechanosensory properties, interneurons (one type of orally directed ascending and three types of anally directed descending interneurons), efferent neurons regulating endocrine cells, inhibitory and excitatory motor neurons, and secretomotor and vasomotor neurons [3,[36][37][38]]"

"enteric neurons with the most correspondent morphology between humans and rodents are classified into two principal types: Dogiel type I neurons have many club-shaped processes and a single long, slender process, whereas Dogiel type II neurons are multipolar and have many long, smooth processes. Type I includes motor neurons consisting of two distinct classes: nitrergic inhibitory and excitatory cholinergic neurons. Each subtype encompasses approximately 5-10% of all enteric neurons within the myenteric plexus. IPANs are thought to be identified with Dogiel type II morphology, with a large, smooth cell body, no dendrites, and several long, uniformly branching axons."

(Nikolla et al., 2025)

The Enteric Neuronal Circuitry: A Key Ignored Player in Nutrient Sensing Along the Gut–Brain Axis

"The ENS is comprised of the myenteric (Auerbach) plexus, nestled between the longitudinal and circular smooth muscle layers, and the submucosal (Meissner) plexus [1], embedded in the submucosal layer of the small intestine that collectively maintain gut homeostasis (Rubin et al., 2009). The myenteric plexus consists of myenteric ganglia of varying sizes. The neurons of the myenteric plexus innervate the muscularis externa and neighboring myenteric neurons (Gabella, 1979). Intrinsic primary afferent neurons (IPANs) have cell bodies in the myenteric plexus (Costa et al., 2000). Additionally, there are intestinofugal neurons (IFANs), a smaller population of myenteric neurons that project to the gallbladder, pancreas, and central nervous system (CNS) (Gabella, 1979). The submucosal plexus has two layers: an outer layer containing motor neurons that project to the smooth muscular layer and an inner layer that innervates the muscularis mucosae, defining the boundary between the mucosa and submucosa (Nezami et al., 2010)"

"Enteric neurons are typically identified and categorized based on morphology, location, chemistry, projections, and functions (Table 1). Historically, neurons have been classified as Dogiel type I or type II across a broad spectrum of species, with type III being the least understood type in humans [8]."

(Brehmer, 2021)

Classification of human enteric neurons

"both animal and human enteric neurons are usually divided into only two morphological subpopulations, "Dogiel type II" neurons (with several long processes) and "Dogiel type I" neurons (with several short processes). This implies no more than the distinction of intrinsic primary afferent from all other enteric neurons."

(Wang et al., 2025)

Dietary Modulation of the Enteric Nervous System: From Molecular Mechanisms to Therapeutic Applications

"Neurons within the ENS are categorized according to various characteristics, such as morphology, neurotransmitter expression, cell surface receptors, electrophysiological properties, connectivity patterns, their specific location within the intestinal wall, and their functional roles. Based on the morphological characteristics of ENS neurons, particularly the shape and length of their dendrites, Dogiel classified these neurons into types Dogiel I-III [28]. With the advent of advanced technologies, notably the application of single-cell RNA sequencing, a more intricate heterogeneity of neurons within the ENS has been uncovered. Through single-nucleus analysis of the human intestine, researchers have identified 436,202 nuclei and successfully isolated 1445 neurons [29]. The functionality of the ENS is dependent on the coordinated action of diverse neurotransmitters, which can be categorized according to their chemical properties and functions. Among cholinergic neurotransmitters, acetylcholine (ACh) is the most prevalent and plays a pivotal role in the ENS [22]"

"Monoaminergic neurotransmitters: Serotonin is a crucial monoamine neurotransmitter with dual roles within the ENS."

(Caillaud et al., 2023)

A functional network of highly pure enteric neurons in a dish

"Several subtypes of enteric neurons have been identified, based on their morphology and neurochemical phenotypes. The morphological criteria concern the size of the cell body, presence of dendrites, and axonal architecture"

"Three distinct morphological classes of transfected neurons were observed. The first class describes monoaxonal neurons with a small cell body diameter ([illegible] 20µm)"

"This first class, predominant, represent 94.7 ± 0.9% of transfected neurons and can be divided in 2 sub-types. Thus, we distinguish

> *neurons with a long axonal process with sparse and short ramifications"*
>
> *"or a medium-length axon with numerous ramifications along the axonal process"*
>
> *"The second class (2.5 ± 0.5% of transfected neurons) is composed of multiaxonal neurons characterized by >2 axons of medium to short length and a large-diameter cell body (40-45 µm)"*
>
> *"Finally, a third and intermediate class was observed, with neurons exhibiting two axons and a small-diameter cell body (□ 20µm) (2.8 ± 0.5% of transfected neurons)"*
>
> *"Interestingly, the first two morphological classes described above are reminiscent of types I and II of Dogiel's classification of enteric neurons, respectively"*
>
> *"the morphological analysis of the dendrites showed that at D3, 23.2 ± 4.5% of them had a lamellar morphology"*
>
> *"52.1 ± 5.5% a filamentous morphology"*
>
> *"and 24.7 ± 1.7% a combination of both morphologies"*
>
> *"Based on neurochemical criteria, the most represented neurons in the ENS are nitrergic and cholinergic neurons"*
>
> *"We found that 20.3 ± 2.6% of total neurons, as determined by Hu staining, were nNOSpositive neurons and 17 ± 1.9% were ChAT-positive neurons"*

(Zhao et al., 2023)

The influence of Nav1.9 channels on intestinal hyperpathia and dysmotility

> *"Neurons in the ENS are mainly divided into Dogiel type I and Dogiel type II neurons in morphology. The cell bodies of Dogiel type I neurons project multiple short dendrites and a single elongated axon. Dogiel type II neurons have smooth cell bodies and project multiple axons, but no dendrites or only a few dendrites [70]. In addition, ENS neurons are divided into sensory neurons, interneurons and motor neurons in terms of functional characteristics [71], and AH-type neurons and S-type neurons in terms of electrophysiological characteristics [72]. AH-type neurons are characterized by the larger action potentials with an infection on the falling phase and long afterhyperpolarizing potential (AHP;>2 s) that follows. S-type neurons are typified by the brief action potentials without slow AHPs, and they present fast excitatory postsynaptic potentials (EPSPs) [68]. The intrinsic sensory neurons/ intrinsic primary afferent neurons (IPANs) of the ENS generally have the characteristics of Dogiel II neurons in morphology, and belong to AH neurons in electrophysiological characteristics [67,73,74]. The ENS comprises the myenteric plexus and the submucosal plexus. The myenteric plexus is mainly responsible for the regulation of intestinal movements. The submucosal plexus is involved in the secretion of water and electrolytes, as well as the neural regulation of intestinal blood flow [68]."*

Functional Classifications of Enteric Neurons

Enteric neurons are functionally classified into five major types based on their roles in gut control: intrinsic primary afferent neurons (IPANs) that detect stimuli, interneurons that relay signals, excitatory and inhibitory motor neurons that control muscle activity, and secretomotor/vasodilator neurons that regulate secretion and blood flow. This functional organization enables complete reflex circuits within the gut wall. (16 sources)

The functional classification of enteric neurons represents a fundamental organizational principle that reflects the complex neural circuitry required for autonomous gut function. Five major functional classes have been consistently identified across species: intrinsic primary afferent neurons (IPANs), interneurons, excitatory motor neurons, inhibitory motor neurons, and secretomotor/vasodilator neurons (Ren et al., 2008) (Jakob et al., 2020) (Bosi et al., 2020) (Kuwahara et al., 2020) (Schonkeren et al., 2021) (Jakob et al., 2021) (Natale et al., 2021) (Mogor et al., 2021) (Marinova et al., 2022) (Bubeck et al., 2023) (Majd et al., 2024) (Jamka et al., 2024) (Xu et al., 2025) (Savulescu-Fiedler et al., 2025) (Pradeloux et al., 2025).

Intrinsic primary afferent neurons (IPANs) serve as the sensory component of the ENS, detecting both chemical and mechanical stimuli from the gastrointestinal tract with cell bodies located in both myenteric and submucosal plexuses (Ren et al., 2008). These neurons are distinct from extrinsic primary afferent neurons whose cell bodies reside in dorsal root or nodose ganglia (Ren et al., 2008). IPANs possess multipolar morphology with bidirectional axons that project to the mucosa to respond to stimuli, giving rise to secretomotor, vasodilator, and motor reflexes (Bosi et al., 2020) (Jamka et al., 2024). These sensory neurons form synapses with all other neuronal types, making them central integrators of enteric neural circuits (Ren et al., 2008).

Interneurons function as relay cells that convey signals between neurons, primarily within the myenteric plexus (Ren et al., 2008) (Jamka et al., 2024). There are two major classes: ascending interneurons that transmit information orally (toward the mouth) and descending interneurons that transmit information in the aboral direction (toward the anus) (Ren et al., 2008) (Jamka et al., 2024). The myenteric plexus typically contains one subtype of ascending interneuron and at least three subtypes of descending interneurons, with some variation in the proximal colon (Ren et al., 2008). Ascending interneurons typically use acetylcholine and tachykinins as neurotransmitters, while descending interneurons employ combinations of acetylcholine, nitric oxide, vasoactive intestinal polypeptide, and ATP (Jamka et al., 2024).

Motor neurons are found in both myenteric and submucosal plexuses and control the effector functions of the gut (Ren et al., 2008). In the myenteric plexus, there are four functional classes: excitatory and inhibitory motor neurons to both the circular and longitudinal smooth muscle layers (Ren et al., 2008) (Bubeck et al., 2023). Excitatory motor neurons are primarily cholinergic, while inhibitory motor neurons to the circular muscle are characterized by nitric oxide synthase and vasoactive intestinal polypeptide immunoreactivity and use ATP as a co-neurotransmitter (Ren et al., 2008) (Pradeloux et al., 2025).

Secretomotor and vasodilator neurons represent specialized motor neurons that control secretory epithelium and blood vessels rather than smooth muscle (Ren et al., 2008) (Kuwahara et al., 2020). In the submucosal plexus, there are two types of excitatory secretomotor neurons projecting to the epithelium and at least one type of vasodilator neuron to submucosal arterioles (Ren et al., 2008). Non-cholinergic secretomotor/vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP, accounting for approximately 45% of submucosal neurons (Ren et al., 2008).

Additional specialized neuronal types include **intestinofugal neurons** that send projections to prevertebral ganglia, providing a pathway for enteric influence on sympathetic function (Ren et al., 2008) (Mogor et al., 2021). Some classifications also recognize neurons that innervate enteroendocrine cells as a distinct motor neuron subclass (Marinova et al., 2022) (Xu et al., 2025). Modern single-cell sequencing studies have provided molecular validation of these functional classifications, revealing that enteric neurons can be broadly grouped based on their primary neurotransmitters - nitric oxide synthase 1 (NOS1) expressing neurons and choline acetyltransferase (ChAT) expressing neurons - with further subdivision based on gene expression patterns that correspond to the classical functional categories (Jakob et al., 2020) (Jakob et al., 2021) (Zeisel et al., 2018).

Evidence

(Ren et al., 2008)

Purinergic receptors and synaptic transmission in enteric neurons

> *"There are enteric neurons that are sensitive to chemical or mechanical cues from the gastrointestinal tract with cell bodies in the myenteric and submucous plexes. They are properly referred to as intrinsic primary afferent neurons (IPANs) and are separate from the extrinsic primary afferent neurons whose cell bodies are in dorsal root or nodose ganglia (Furness et al., 1998). In this review we will simply refer to them as sensory neurons. Many enteric sensory neurons have "AH"-type electrophysiological characteristics which are defined as having an action potential afterhyperpolarization lasting >1 s (Hirst et al., 1974)(Bornstein et al., 1994)(Mao et al., 2006). Most sensory neurons have smooth cell bodies and are multipolar with projections that run to the mucosal epithelium, between the myenteric plexus and the submucosal plexus, and to their own and nearby ganglia (Pompolo et al., 1988). They form synapses with all other neuronal types"*

> *"Enteric interneurons are primarily found in the myenteric plexus where there is generally one subtype of interneuron (two types in proximal colon) with an ascending projection and at least three subtypes of interneuron with a descending projection. There are also a small number of intestinofugal neurons that send a projection to prevertebral ganglia (Furness et al., 2002). All interneurons have a single axon, but there is a mix of dendritic morphology. Interneurons are characterized electrophysiologically as "S" neurons"*

> *"In the myenteric plexus, ATP is utilized as a transmitter or co-transmitter by at least one type of descending interneuron during local reflexes (Bornstein, 2008)[17]. These descending interneurons are immunoreactive for nitric oxide synthase (NOS) and vasoactive intestinal polypeptide (VIP) and have Dogiel type I dendrites"*

> *"Enteric motor neurons are found in the myenteric and submucous plexes. They have a single axon and short dendrites with Dogiel*

type I morphology. In the myenteric plexus, there are four functional classes of motor neuron to the smooth muscle layers of the intestine (Furness et al., 2002)(Brookes, 2001): inhibitory or excitatory motor neurons to either the circular or longitudinal smooth muscle layers. All motor neurons have S-type properties"

"In the myenteric plexus, inhibitory motor neurons to the circular muscle are immunoreactive for NOS and VIP and use ATP as a coneurotransmitter (Furness et al., 2002)(Costa et al., 1996)(Crist et al., 1992). In the submucous plexus, there are two types of excitatory secretomotor neurons going to the epithelium and at least one type of vasodilator neuron to the submucous arterioles. Noncholinergic secretomotor/ vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP. These neurons account for ~45% of submucosal neurons, and they may utilize ATP as a coneurotransmitter (Monro et al., 2004)(Hu et al., 2003)."

(Jakob et al., 2020)

Neuro-Immune Circuits Regulate Immune Responses in Tissues and Organ Homeostasis

"The ENS is organized in plexuses throughout the intestine composed of neurons whose cell bodies lie within the intestinal wall (Figure 2)"

"The myenteric plexus (Auerbach plexus) lies between the longitudinal and circular muscle layer in the intestinal wall whereas the submucosal plexus (Meissner plexus) forms a network within the submucosal layer (Figure 2)"

"Neurons in the ENS can be categorized based on their anatomy, function and neurotransmitter signature. Up to 20 functional classes of neurons can be identified in the guinea pig. Functionally and phenotypically, several types of enteric neurons are distinguished and can be further sub-classified: Excitatory neurons innervating intestinal muscles, inhibitory neurons innervating intestinal muscles (to circular and longitudinal muscles, respectively), secretomotor and vasodilator neurons, secretomotor neurons without vasodilator activity and neurons to enteroendocrine cells, sensory intrinsic primary afferent neurons, ascending and descending interneurons and intestinofugal neurons (Furness, 2000)(Furness, 2006)(Furness, 2008). Single-cell sequencing experiments revealed nine clusters of enteric neurons in mice, which can be classified based on the two neurotransmitters nitric oxide (NO) (Nos1 expression, cluster 1-3) and acetylcholine (Chat expression, cluster 4-9) (Zeisel et al., 2018). Structurally, Nos1 + neurons are preferentially type I neurons whereas among Chat + neurons type II neurons are overrepresented (34)."

(Bosi et al., 2020)

Tryptophan Metabolites Along the Microbiota-Gut-Brain Axis: An Interkingdom Communication System Influencing the Gut in Health and Disease

"The ENS is characterized by 3 major ganglionated plexuses, the submucosal, myenteric, and subserous plexus, separated by interconnecting fibre strands. nteric neurons are classified according to their morphology, neurochemical coding, projections to targets, and functional roles into 5 major types: intrinsic primary afferent neurons (IPAN), interneurons, excitatory and inhibitory motor neurons, and secretomotor neurons. In the submucosal and myenteric plexus, IPANs possess large-cell bodies and bidirectional axons projecting to the mucosa to respond to chemical and mechanical stimuli giving rise to secretomotor, vasodilator, and motor reflexes. This neuronal subtype is highly conserved in different animal species and can be identified by the presence of hyperpolarization-activated cation current and post-action potential and by the expression of different neurotransmitters such as substance P, acetylcholine, and calcitonin gene-related peptide and 5-HT."

(Kuwahara et al., 2020)

Microbiota-gut-brain axis: enteroendocrine cells and the enteric nervous system form an interface between the microbiota and the central nervous system.

"Myenteric and submucosal neurons are composed of discrete populations of neurons that can be classified based on their function and morphology. These include intrinsic primary afferent neurons (IPANs) that allow them to regulate GI motility and secretion without CNS input, motor neurons (muscle, secretomotor and vasodilator neurons) and interneurons"

"Neurons in the ENS are also divided into two subtypes based on their electrophysiological properties, which correlate with their morphologies (Furness, 2012). AH (after hyperpolarizing) neurons have multiple long processes and a large oval soma (AH/ Dogiel Type II neurons) and S (synaptic) neurons (Brookes et al., 1995)(Clerc et al., 1997)(Nurgali et al., 2004)). Morphology of S neuron is flattened, slightly elongated with stellate or angular forms. The characteristic belongs to Dogiel Type I neurons. S neurons function as muscular motors, secretomotors and interneurons. On the other hand, AH neurons are chemo-and mechanosensitive IPANs."

(Schonkeren et al., 2021)

The Emerging Role of Nerves and Glia in Colorectal Cancer

"The major neuronal types in the enteric nervous system are intrinsic primary afferent neurons (sensory neurons with Dogiel type II morphology that project to the mucosa, black), descending (purple) and ascending (green) interneurons, secretomotor and vasodilatator neurons (red), excitatory (blue) and inhibitory (orange) motorneurons, and intestinofugal neurons (pink)."

(Jakob et al., 2021)

An Integrated View on Neuronal Subsets in the Peripheral Nervous System and Their Role in Immunoregulation

"The somas of enteric neurons mainly cluster in two anatomically distinct but strongly interconnected ganglionated plexuses (10). The outer, myenteric plexus (Auerbach Plexus) lies within the longitudinal and the circular muscle layer, and the inner, submucosal plexus (Meissner Plexus) is located below the muscle layers (Jakob et al., 2020)"

"The intrinsic micro-circuitry regulation of intestinal function consists of five neuronal classes with distinct functional specializations. Sensory neurons, also referred to as intrinsic primary afferent neurons (IPANs), detect chemical and physical alterations in the intestine and transmit the signal via interneurons to excitatory motor neurons, inhibitory motor neurons or secretomotor/vasodilator neurons"

"Single-cell sequencing of the ENS using fluorescenceactivated sorting of Wnt1-Cre; R26Tomato mice revealed 1105 enteric neurons (with ~3066 genes detected on average) and eventually found nine clusters of neurons in the muscular sheet of the intestine (Zeisel et al., 2018). Based on neuro-transmitters nitric oxide synthase 1 (NOS1) and choline acetyltransferase (ChAT), two main groups can be classified, NOS1 + neurons and ChAT + neurons. The authors assigned three distinct neuronal clusters of nitrergic neurons, which comprise inhibitory motor neurons and secretomotor/vasodilator neurons (Table 1). Further, 6 clusters of cholinergic neurons, which express ChAT and the solute carrier family 5 member 7 (Slc5a7, gene encodes for an ion transporter) could be distinguished (Zeisel et al., 2018). ChAT + neurons are further subdivided into excitatory motor neurons, sensory neurons and interneurons according to their gene expression (Table 1)"

"Using the pan-neuronal Baf53b-cre deleter mice crossed to R26R-tomato to enable sort-purification of enteric neurons from the myenteric plexus of the small intestine combined with singlecell RNA-sequencing, 4,892 high-quality enteric neurons have been analyzed and 12 classes of neurons could be identified based on their gene expression (13)."

(Natale et al., 2021)

[The Baseline Structure of the Enteric Nervous System and Its Role in Parkinson's Disease]

"Intrinsic neurons of the ENS can be classified according to different criteria and approaches: morphological features (neuron size and distribution, and axonal projections, by means of light and electron microscopy), electrophysiological properties, and biochemical markers (neurotransmitter content and receptors, by means of histochemistry, immunohistochemistry, and transcript analysis)"

"Several types of intrinsic neurons that belong to the ENS have been described (Figure 3). These include intrinsic primary afferent neurons (IPANs) possessing chemo-and mechanosensory properties, interneurons (one type of orally directed ascending and three types of anally directed descending interneurons), efferent neurons regulating endocrine cells, inhibitory and excitatory motor neurons, and secretomotor and vasomotor neurons [3,[36][37][38]]"

"enteric neurons with the most correspondent morphology between humans and rodents are classified into two principal types: Dogiel type I neurons have many club-shaped processes and a single long, slender process, whereas Dogiel type II neurons are multipolar and have many long, smooth processes. Type I includes motor neurons consisting of two distinct classes: nitrergic inhibitory and excitatory cholinergic neurons. Each subtype encompasses approximately 5-10% of all enteric neurons within the myenteric plexus. IPANs are thought to be identified with Dogiel type II morphology, with a large, smooth cell body, no dendrites, and several long, uniformly branching axons."

(Mogor et al., 2021)

[The Enteric Nervous System and the Microenvironment of the Gut: The Translational Aspects of the Microbiome-Gut-Brain Axis]

"The oldest classification of enteric neurons was based on morphology (Brehmer et al., 1999); this usually distinguished two morphological subpopulations: 'Dogiel type II' neurons with several long processes and 'Dogiel type I' neurons with numerous short processes"

"Electrophysiological classification of enteric neurons is also useful in defined circumstances. The first experiments to capture their electrical properties (which later provided the basis for classification) were recorded in the 1970s (Hirst et al., 1974)(Nishi et al., 1973)[32], where S (synaptic)-and AH (after-hyperpolarizing)-type neurons were identified"

"Other functional subsets of enteric neurons can be inferred based on their axonal projection patterns, their synthesizing enzymes, calcium-binding proteins, cytoskeletal proteins and expression of primary transmitters (Fantaguzzi et al., 2009)(Qu et al., 2008). Nowadays, the enteric neurons are mostly divided into four main classes: motoneurons, intrinsic primary afferent neurons (IPANs), interneurons and intestinofugal neurons"

"Altogether, 14 different subtypes had been described previously (summarized in Table 2), but in a recent study, a new intrinsic afferent neuron subtype was also identified (Nestor-Kalinoski et al., 2021). The substances that act as neurotransmitters in the central nervous system are also found in the enteric nervous system, so that in addition to the classical neurotransmitters, several neuropeptides, and other neurotransmitters (adenosine triphosphate (ATP) and nitric oxide (NO)) also play an important role."

(Marinova et al., 2022)

[ACETYLCHOLINE AND ITS SYNTHESIZING ENZYME CHOLINE ACETYLTRANSPHERASE IN THE ENTERIC NERVOUS SYSTEM]

"According to their morphological and functional properties, enteric neurons are classified into motor neurons, sensory neurons, and interneurons. There are three types of motor neurons in the ENS: muscle motor neurons, secretomotor/visceromotor neurons and neurons that innervate entero-endocrine cells. Muscle motor neurons innervate the longitudinal, circular muscles and muscular is mucosae through the digestive tract. Excitatory and inhibitory circular muscle motor neurons have a Dogiel type I shape. They have a stellar-shaped cell body with many short dendrites which divide several times into short, flat branches. Secretomotor and vasomotor neurons are two small classes of neurons which are more frequent in the submucosal plexus. Dogiel supposed secretomotor neurons to be type II neurons [3,5,6]"

"The second type of sensory neurons are the intrinsic primary afferent neurons (IPANs). The bodies of these neurons are within the gut wall in the submucosal and myenteric plexus. The IPANs are multipolar cells with Dogiel II characteristics [12]"

"Interneurons are ascending (orally directed) interneurons and several classes of descending (anally directed) neurons. Most of the ascending enteric interneurons belong to the Dogiel type I neurons"

"Most of the interneurons are from the descending type."

(Bubeck et al., 2023)

Guardians of the gut: influence of the enteric nervous system on the intestinal epithelial barrier

"excitatory (PEMN/eMN) and inhibitory (PIMN/iMN) motor neurons innervate both the circular and longitudinal muscles and are connected by interneurons (PIN/IN). Sensory neurons (PSN) extend through the circular muscle layer, innervating the submucosal plexus as well as the epithelial layer. With the advent of single cell transcriptomic technologies four major studies have reported the single cell transcriptomes of the human and mouse ENS. Following clustering and annotation of neuronal function, combined with cluster-labelling gene expression patterns, there have emerged seven distinct subtypes"

"The submucosal plexus, located beneath the circular muscle layer, primarily innervates the submucosal layer and the surrounding crypts of the enteric epithelium. It receives input from primary afferent neurons (IPAN) or secretomotor/vasodilator neurons (PSVN) and exhibits a more compact network with fewer cell bodies compared to the myenteric plexus. The main function of the submucosal plexus is to regulate secretory activity (Timmermans et al., 1997)."

(Majd et al., 2024)

A call for a unified and multimodal definition of cellular identity in the enteric nervous system

"For decades, enteric neuron subtype identity has largely been defined based on morphological, immunohistochemical and functional features of neurons within the gut, including neuronal projection localization, neurochemistry, and electrophysiological properties. Investigations conducted on tissues isolated from model organisms, such as the guinea pig, mouse and rat 1 have significantly progressed our understanding of the neurochemical and functional complexity of enteric neuron subtypes and have identified functional units within the ENS circuitry. These include intrinsic primary afferent neurons (IPANs or sensory neurons, SNs), ascending and descending interneurons (IN), excitatory and inhibitory motor neurons (EMNs and IMNs), and secretomotor and/or vasodilator neurons (SVNs) 1 . Despite the fact that many known neurotransmitters and neuropeptides are present in the ENS, only a handful are consistently used as markers to identify major cell types in the ENS. Some examples include nitric oxide synthase (NOS1) for IMNs and acetylcholine for EMNs as well as CALCA/B for IPANs 2 . While it is impressive how large swaths of ENS function are coordinated by these neurotransmitters and neuropeptides, it also raises the question of whether yet-to-be-determined factors confer cell-type specific functional properties 3 . These features might include secreted peptides and proteins, electrophysiological features mediated by ion channels and transporters, metabolic profiles, and specialized capabilities for cell-cell interaction or synapse formation. To accurately understand the identity of individual cell types, it is also crucial to consider the contextual information in which they operate, including their position in the gut. Histochemical and functional assays have identified differences along the length of the GI tract, as well as between species, highlighting the need for more systematic efforts towards comprehensive mapping and profiling of the ENS 1 ."

(Jamka et al., 2024)

Mechanisms of enteric neuropathy in diverse contexts of gastrointestinal dysfunction

"Enteric neurons are broadly grouped into four major functional classes: intrinsic primary afferent neurons (IPANs), motor neurons, interneurons, and intestinofugal neurons. IPANs are present in the myenteric and submucosal plexuses and are multipolar cells with a Dogiel type II morphology. These neurons have at least one process that branches into the mucosa and multiple other projections that extend from the cell body to synapse with other neurons and enteric glia within the plexus"

"Enteric interneurons convey signals between neurons in the myenteric plexus and are primarily unipolar relay cells with Dogiel type I morphology. There are two major classes of interneurons; ascending which transmits information orally and descending which transmits information in the aboral direction"

"ascending interneurons typically use acetylcholine and tachykinins as neurotransmitters while descending interneurons use combinations of acetylcholine, nitric oxide (NO), vasoactive intestinal peptide (VIP), and adenosine triphosphate (ATP) as neurotransmitters"

(Xu et al., 2025)

Tryptophan and Its Metabolite Serotonin Impact Metabolic and Mental Disorders via the Brain–Gut–Microbiome Axis: A Focus on Sex Differences

> *"The ENS consists of three anatomical ganglionated plexuses of submucosal, myenteric, and subserous plexuses, interconnected by fibers (O'Mahony et al., 2015). Five major types of enteric neurons, based on their morphological, neurochemical, and functional characteristics, are intrinsic primary afferent neurons, interneurons, and excitatory, inhibitory, and secreto-motor neurons [38] (Furness, 2012). These neurons form a bidirectional network, with afferent neurons in the submucosal and myenteric plexuses responding to chemical and mechanical stimuli and regulating vasodilation and blood flow, motility, mucosal transport, secretion, and nutrient absorption (Furness et al., 2004)."*

(Savulescu-Fiedler et al., 2025)

Rewiring the Brain Through the Gut: Insights into Microbiota–Nervous System Interactions

> *"The ENS is structured with two main plexuses represented by the submucous plexus (Meissner) and the myenteric plexus (Auerbach)"*
>
> *"Both plexuses consist of three neuron types: the enteric intrinsic primary afferent neurons (IPANs), the motor neurons or efferent neurons, and the association neurons"*
>
> *"IPANs represent the most vulnerable ENS neuronal population, given the fact that they extend to the lamina propria of the intestine (Benarroch, 2007)(Furness et al., 2004)"*
>
> *"The enteric peristalsis is controlled by the motor neurons arising from the myenteric plexus, while absorption and secretion are regulated by the neurons emerging from the submucosal plexus (Furness et al., 2014)"*
>
> *"The main classes of enteric neurons are represented by neurons that receive fast synaptic input, known as "S" neurons, and neurons with prolonged post-hyperpolarization, referred to as "AH" neurons (Natale et al., 2021). IPANs belong to "AH" neurons, whereas motor and interneurons are "S" neurons, which are relatively depolarized at rest [69]."*

(Pradeloux et al., 2025)

Impact of aging on the central and enteric nervous system in a Parkinson's disease mouse model

> *"Functionally, enteric neurons are typically classified into five major categories based on their neurotransmitter profiles:*
>
> *(1) excitatory motor neurons (cholinergic), (2) inhibitory motor neurons (nitrergic), (3) sensory neurons, (4) excitatory and inhibitory interneurons, and (5) secretomotor/vasodilator neurons (Natale et al., 2021). The cholinergic neurons make up the majority, at least 70% of myenteric neurons (Anlauf et al., 2003)"*
>
> *"Those neurons, constituting less than 1% of all enteric cells, are organized into ganglionic plexuses-primarily the myenteric (MyP) and submucosal (SMP) plexuses-or into non-ganglionic nerve fiber bundles (Furness et al., 1980)(Brudek, 2019)"*
>
> *"DAergic neurons represent a relatively small but functionally significant subset of the ENS. In mice, they account for approximately 10-13% of enteric neurons, while in humans, this proportion can reach up to 20%, with a distribution that follows an oral-to-aboral gradient (Anlauf et al., 2003)(Li et al., 2003). These neurons are particularly enriched in the upper gastrointestinal tract, where they constitute 14-20% of the neuronal population, but their prevalence drops to 1-6% in the distal small intestine. DAergic enteric neurons express key dopaminergic markers, including tyrosine hydroxylase (TH), dopamine transporter (DAT), and all five dopamine receptor subtypes (D1-D5)"*
>
> *"Expression of the dopaminergic phenotype in the mouse ENS is variable depending on the region of the gut, TH transcripts being more prominent in the stomach and ileum compared to the colon (Chalazonitis et al., 2022)."*

(Zeisel et al., 2018)

Molecular Architecture of the Mouse Nervous System

> *"The mammalian nervous system executes complex behaviors controlled by specialised, precisely positioned and interacting cell types. Here, we used RNA sequencing of half a million single cells to create a detailed census of cell types in the mouse nervous system. We mapped cell types spatially and derived a hierarchical, data-driven taxonomy. Neurons were the most diverse, and were grouped by developmental anatomical units, and by the expression of neurotransmitters and neuropeptides. Neuronal diversity was driven by genes encoding cell identity, synaptic connectivity, neurotransmission and membrane conductance. We discovered several distinct, regionally restricted, astrocytes types, which obeyed developmental boundaries and correlated with the spatial distribution of key glutamate and glycine neurotransmitters. In contrast, oligodendrocytes showed a loss of regional identity, followed by a secondary diversification. The resource presented here lays a solid foundation for understanding the molecular architecture of the mammalian nervous system, and enables genetic manipulation of specific cell types."*

IPAN Functional Subtypes and Properties

Intrinsic primary afferent neurons (IPANs) are polymodal sensory neurons with distinct electrophysiological, morphological, and molecular properties that enable them to detect chemical and mechanical stimuli throughout the gut wall. Recent single-cell sequencing studies have revealed multiple molecular subtypes of IPANs with specialized neurotransmitter profiles and anatomical distributions. (9 sources)

Intrinsic primary afferent neurons represent a specialized class of enteric sensory neurons with distinctive properties that enable their function as the primary detectors of gut stimuli. IPANs are characterized by their **AH-type electrophysiological properties**, defined by action potentials with afterhyperpolarization lasting greater than 1 second, which distinguishes them from S-type neurons found in motor neurons and interneurons (Ren et al., 2008) (Fung et al., 2020). This prolonged afterhyperpolarization enables IPANs to set levels of excitability through somatic gating of activity within enteric networks (Fung et al., 2020).

Morphologically, IPANs correspond to Dogiel Type II neurons with large, round or oval cell bodies and multiple long processes that project bidirectionally - to the mucosa to detect stimuli and to other ganglia to initiate reflex responses (Bosi et al., 2020) (Kuwahara et al., 2020). These multipolar neurons have extensive projections that form synapses with all other neuronal types, making them central integrators of enteric circuits (Ren et al., 2008). IPANs are typically polymodal, possessing both chemosensory and mechanosensory properties that allow them to respond to chemical stimuli, mucosal deformation, and gastrointestinal muscle tension (Fung et al., 2020) (Magalhaes et al., 2021).

Molecular markers that characterize IPANs include substance P, acetylcholine, calcitonin gene-related peptide (CGRP), and serotonin (Bosi et al., 2020). Recent single-cell RNA sequencing studies have revealed significant **molecular diversity among IPAN subtypes**. In the mouse myenteric plexus, at least three distinct IPAN classes have been identified that express different combinations of established markers including Calca/Calcb/Nfem/Calb1/Calb2, with clear separation based on selective expression of neuromedin U (Nmu), urocortin-3/cholecystokinin (Ucn-3/Cck), or neurexophilin-2 (Nxph2) (Morarach et al., 2020). Additional molecular markers that distinguish IPAN subtypes include Calcb and Nmu expression, along with cadherin 6 (Cdh6) as a novel marker (Gomez-Frittelli et al., 2025) (Gomez-Frittelli et al., 2024).

Anatomical distribution of IPANs shows important regional specialization. While IPANs are present in both myenteric and submucosal plexuses, only submucosal IPANs are sensitive to subtle mechanical stimulation of the mucosa (Fung et al., 2020). In the myenteric plexus, IPANs represent approximately 30% of neurons, while in the submucosal plexus they comprise about 14% of the neuronal population (Magalhaes et al., 2021). The somata are predominantly localized within the myenteric plexus, with axonal terminals extensively projecting to the mucosal layer and adjacent plexuses to form dense sensory networks (Sun et al., 2025).

Functional properties of IPANs include their ability to generate and sustain responses to subthreshold stimuli through their characteristic low-threshold action potentials and prominent afterhyperpolarization potentials (Sun et al., 2025). Upon stimulation, IPANs release neurosignaling molecules such as tachykinins and CGRP, initiating secretomotor, vasodilator, and motor reflexes that coordinate gut function (Sun et al., 2025) (Bosi et al., 2020). This IPAN subtype diversity appears to be highly conserved across different animal species, suggesting fundamental importance for enteric nervous system function (Bosi et al., 2020).

Evidence

(Ren et al., 2008)

Purinergic receptors and synaptic transmission in enteric neurons

> *"There are enteric neurons that are sensitive to chemical or mechanical cues from the gastrointestinal tract with cell bodies in the myenteric and submucous plexes. They are properly referred to as intrinsic primary afferent neurons (IPANs) and are separate from the extrinsic primary afferent neurons whose cell bodies are in dorsal root or nodose ganglia (Furness et al., 1998). In this review we will simply refer to them as sensory neurons. Many enteric sensory neurons have "AH"-type electrophysiological characteristics which are defined as having an action potential afterhyperpolarization lasting >1 s (Hirst et al., 1974)(Bornstein et al., 1994)(Mao et*

al., 2006). Most sensory neurons have smooth cell bodies and are multipolar with projections that run to the mucosal epithelium, between the myenteric plexus and the submucosal plexus, and to their own and nearby ganglia (Pompolo et al., 1988). They form synapses with all other neuronal types"

"Enteric interneurons are primarily found in the myenteric plexus where there is generally one subtype of interneuron (two types in proximal colon) with an ascending projection and at least three subtypes of interneuron with a descending projection. There are also a small number of intestinofugal neurons that send a projection to prevertebral ganglia (Furness et al., 2002). All interneurons have a single axon, but there is a mix of dendritic morphology. Interneurons are characterized electrophysiologically as "S" neurons"

"In the myenteric plexus, ATP is utilized as a transmitter or co-transmitter by at least one type of descending interneuron during local reflexes (Bornstein, 2008)[17]. These descending interneurons are immunoreactive for nitric oxide synthase (NOS) and vasoactive intestinal polypeptide (VIP) and have Dogiel type I dendrites"

"Enteric motor neurons are found in the myenteric and submucous plexes. They have a single axon and short dendrites with Dogiel type I morphology. In the myenteric plexus, there are four functional classes of motor neuron to the smooth muscle layers of the intestine (Furness et al., 2002)(Brookes, 2001): inhibitory or excitatory motor neurons to either the circular or longitudinal smooth muscle layers. All motor neurons have S-type properties"

"In the myenteric plexus, inhibitory motor neurons to the circular muscle are immunoreactive for NOS and VIP and use ATP as a coneurotransmitter (Furness et al., 2002)(Costa et al., 1996)(Crist et al., 1992). In the submucous plexus, there are two types of excitatory secretomotor neurons going to the epithelium and at least one type of vasodilator neuron to the submucous arterioles. Noncholinergic secretomotor/ vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP. These neurons account for ~45% of submucosal neurons, and they may utilize ATP as a coneurotransmitter (Monro et al., 2004)(Hu et al., 2003)."

(Fung et al., 2020)

Functional circuits and signal processing in the enteric nervous system

"Dogiel published illustrations of three morphological subtypes of neurons in the ENS, defined as Dogiel Types I, II, and III [14]. It was not until 1974 that Hirst and colleagues then reported on their electrophysiological properties and classified them into AH-and S-type neurons (Hirst et al., 1974). AH neurons are typified by their larger action potentials (APs) with an inflection on the falling phase and long afterhyperpolarizing potential (AHP; > 2 s) that follows. By comparison, S neurons are characterized by their brief APs that lack slow AHPs, and they display fast excitatory postsynaptic potentials (EPSPs)"

"Costa, Furness, and colleagues conducted an extensive series of immunohistochemical and electron microscopy studies to group enteric neurons based on their distinct neurochemistry, projection patterns, and ultrastructure (Costa et al., 1996)(Costa et al., 1983)(Costa et al., 1981)(Costa et al., 1992)(Furness et al., 2004)(Furness et al., 1982)(Furness et al., 2004)[23][24][25][26][27][28][29]"

"immunostaining for the expression of primary transmitters or their synthesizing enzymes, and cytoskeletal and calcium-binding proteins"

"studies using various single-cell RNA-sequencing methods optimized for the ENS are now emerging, allowing for extensive gene expression profiling of functional molecular constituents of enteric neurons such as ion channels, transmitters and their synthesizing enzymes, as well as receptors [35,36]. This methodology based on statistical clustering algorithms subdivided myenteric neurons into nine molecularly defined subpopulations. By comparison, ten different classes of myenteric neurons were identified from integrated anatomical, neurochemical, functional, and pharmacological analysis (Furness et al., 2014)"

"Intrinsic sensory neurons (or IPANs) are typically polymodal, possessing chemosensory and mechanosensory properties, and have Dogiel Type II morphology [40]. Furthermore, they are associated with AH-type characteristics where their AHP enables them to set a level of excitability through somatic gating of activity within the network [41]"

"singlecell sequencing data provides evidence for several subclasses [35]36]. One clear division is whether they are situated in the submucosal or myenteric plexus. Only the former subset are sensitive to subtle mechanical stimulation of the mucosa"

(Bosi et al., 2020)

Tryptophan Metabolites Along the Microbiota-Gut-Brain Axis: An Interkingdom Communication System Influencing the Gut in Health and Disease

"The ENS is characterized by 3 major ganglionated plexuses, the submucosal, myenteric, and subserous plexus, separated by interconnecting fibre strands. nteric neurons are classified according to their morphology, neurochemical coding, projections to targets, and functional roles into 5 major types: intrinsic primary afferent neurons (IPAN), interneurons, excitatory and inhibitory motor neurons, and secretomotor neurons. In the submucosal and myenteric plexus, IPANs possess large-cell bodies and bidirectional axons projecting to the mucosa to respond to chemical and mechanical stimuli giving rise to secretomotor, vasodilator, and motor reflexes. This neuronal subtype is highly conserved in different animal species and can be identified by the presence of hyperpolarization-activated cation current and post-action potential and by the expression of different neurotransmitters such as substance P, acetylcholine, and calcitonin gene-related peptide and 5-HT."

(Kuwahara et al., 2020)

[Microbiota-gut-brain axis: enteroendocrine cells and the enteric nervous system form an interface between the microbiota and the central nervous system.]

> *"Myenteric and submucosal neurons are composed of discrete populations of neurons that can be classified based on their function and morphology. These include intrinsic primary afferent neurons (IPANs) that allow them to regulate GI motility and secretion without CNS input, motor neurons (muscle, secretomotor and vasodilator neurons) and interneurons"*

> *"Neurons in the ENS are also divided into two subtypes based on their electrophysiological properties, which correlate with their morphologies (Furness, 2012). AH (after hyperpolarizing) neurons have multiple long processes and a large oval soma (AH/ Dogiel Type II neurons) and S (synaptic) neurons (Brookes et al., 1995)(Clerc et al., 1997)(Nurgali et al., 2004)). Morphology of S neuron is flattened, slightly elongated with stellate or angular forms. The characteristic belongs to Dogiel Type I neurons. S neurons function as muscular motors, secretomotors and interneurons. On the other hand, AH neurons are chemo-and mechanosensitive IPANs."*

(Magalhaes et al., 2021)

[Enteric nervous system and inflammatory bowel diseases: Correlated impacts and therapeutic approaches through the P2X7 receptor]

> *"As proposed by Aleksandr S. Dogiel in 1899, the morphological classification of enteric neurons can be based on their conformation and dendritic distribution. Dogiel described type I cells as flattened, slightly elongated, with an angled or star-shaped contour, and, as remarkable characteristics, as having only one axon and four to 20 Lamellar dendrites that frequently extend at a short distance from the cell body [31]"*

> *"Type II neurons have large round or oval cell bodies and eccentric nuclei (Pompolo et al., 1988), and the surface is grooved by bundles of neural fibers (Hendriks et al., 1990). The main characteristic of type II neurons is the presence of several axonal processes that are emitted either directly from the cell body (multipolar neuron) or from a single initial process that branches into short subsidiary axons (pseudounipolar neurons) (Furness, 2012)(Song et al., 1994)"*

> *"Additionally, enteric neurons can also be identified as intrinsic primary afferent neurons (IPANs), interneurons, and motor neurons (Furness, 2012), classified into at least 18 subtypes and using more than 30 neurotransmitters in their synapses [28]30]. Of these neurotransmitters, acetylcholine (ACh) and nitric oxide (NO) stand out as the most abundant (Brookes, 2001), as well as adenosine-5'-triphosphate (ATP) (Hansen, 2003), vasoactive intestinal polypeptide (VIP), and substance P (SP) (Gallego et al., 2016)"*

> *"IPANs (classified as Dogiel type II) are recognized for responding to chemical stimuli, mucosal deformation and GI muscle tension, translating these signals into a neural impulse that will trigger a local motor reflex (Costa et al., 2000). Altogether, IPANs represent approximately 14% and 30% of the neurons of the submucosal and myenteric plexuses, respectively"*

> *"The enteric neural circuit is arranged in two plexuses: the submucosal plexus, which in large mammals is present in two individualized levels (outer/inner) and is located in the outer connective tissue layer and the inner mucosal layer, and the myenteric plexus, located between the longitudinal and circular muscle layers (Bayliss et al., 1899)(Furness, 2012)5,28]"*

(Morarach et al., 2020)

[Diversification of molecularly defined myenteric neuron classes revealed by single cell RNA-sequencing]

> *"Meticulous work over the last decades has identified a framework of enteric neurons with sensory (Intrinsic Primary Afferent Neurons, IPANs), motor or interneuron characteristics"*

> *"we define a molecular taxonomy of 12 enteric neuron classes within the myenteric plexus of the mouse small intestine using single-cell RNA sequencing. We present cell–cell communication features and histochemical markers for motor neurons, sensory neurons and interneurons"*

> *"some classes were defined by the selective expression of single genes with likely importance for their neuronal activity. These included ENC5 (somatostatin; Sst), ENC6 (neuromedin U; Nmu), ENC7 (cholecystokinin; Cck and urocortin 3; Ucn3), ENC10 (neuronal differentiation 6; Neurod6) and ENC12 (neurexophilin-2; Nxph2). Other classes were distinguished by combinatorial gene codes"*

> *"ENC1-4 displayed a partly shared gene expression, including Tac1, but differed in their expression of Calb2, Ndufa4l2, Gda, Penk and Fut9"*

> *"ENC8-9 jointly expressed Nos1 and c1ql1, but Npy and Cox8b were significantly lower in ENC9. ENC11 highly expressed Npy, but lacked Nos1"*

> *"ENC1-4 were assigned to excitatory motor neurons (Tac1/Calb2), while ENC8 and 9 matched inhibitory motor neurons (Nos1/Gal/ Vip/Npy). ENC6, 7 and 12 were mapped to IPANs and expressed different combinations of previously defined markers (Calca/Calcb/ Nfem/Calb1/Calb2). These plausible IPANs were clearly separable by their selective expression of Nmu, Ucn-3/Cck or Nxph2."*

(Gomez-Frittelli et al., 2025)

Synaptic cell adhesion molecule Cdh6 identifies a class of sensory neurons with novel functions in colonic motility

> *"Neuronal subtypes in the ENS have been distinguished by their electrophysiological properties, morphology, and expression of characteristic markers, notably neurotransmitters and neuropeptides"*

> *"Cdh6+ neurons demonstrate all other distinguishing classifications of enteric sensory neurons including marker expression of Calcb and Nmu, Dogiel type II morphology and AH-type electrophysiology and IH current"*

> *"These neuronal subtypes have begun to be distinguished morphologically, electrophysiologically, and by marker expression, classically especially of neurotransmitters (Nurgali et al., 2004)(Qu et al., 2008)"*

> *"Cdh6 neurons express IPAN markers Calcb and Nmu. Sparse labeling of individual IPANs reveals they project mainly circumferentially and branch extensively in myenteric ganglia."*

(Gomez-Frittelli et al., 2024)

Synaptic cell adhesion molecule Cdh6 identifies a class of sensory neurons with novel functions in colonic motility

> *"Neuronal subtypes in the ENS have been distinguished by their electrophysiological properties, morphology, and expression of characteristic markers, notably neurotransmitters and neuropeptides"*

> *"Cdh6+ neurons demonstrate all other distinguishing classifications of enteric sensory neurons including marker expression of Calcb and Nmu, Dogiel type II morphology and AH-type electrophysiology and IH current."*

(Sun et al., 2025)

Gut sensory neurons as regulators of neuro-immune-microbial interactions: from molecular mechanisms to precision therapy for IBD/IBS

> *"Morphologically classified as Dogiel type II multipolar neurons, their somata are predominantly localized within the myenteric plexus, with axonal terminals extensively projecting to the mucosal layer and adjacent plexuses, forming a dense sensory network (Spencer et al., 2020) (Fig. 1). Upon mechanical or chemical stimulation, IPANs generate action potentials through depolarization, releasing neurosignaling molecules such as tachykinins and CGRP"*

> *"Characterized as AH-type neurons, IPANs display prominent afterhyperpolarization (AHP) potentials and low-threshold action potentials, allowing sustained responsiveness to subthreshold stimuli [11]."*

Electrophysiological Properties: AH-type vs S-type Neurons

Enteric neurons are classified into two main electrophysiological types: AH-type neurons with prolonged afterhyperpolarization potentials lasting over 1-2 seconds that correspond to sensory IPANs, and S-type neurons with brief action potentials and fast synaptic responses that include motor neurons and interneurons. (10 sources)

The electrophysiological classification of enteric neurons, first established by Hirst and colleagues in 1974, provides a fundamental framework for understanding ENS function based on the distinct electrical properties of different neuronal populations (Fung et al., 2020) (Sayegh et al., 2013). This classification system divides enteric neurons into two primary categories that correlate closely with their morphological and functional characteristics.

AH-type neurons are characterized by larger action potentials with an inflection on the falling phase and a prolonged afterhyperpolarizing potential (AHP) lasting greater than 1-2 seconds (Fung et al., 2020) (Ren et al., 2008) (Zhao et al., 2023). This distinctive afterhyperpolarization enables AH-type neurons to set levels of excitability through somatic gating of activity within enteric networks (Fung et al., 2020). These neurons also display prominent slow excitatory synaptic inputs and lack fast excitatory synaptic inputs (Kuwahara et al., 2020). The prolonged AHP allows sustained responsiveness to subthreshold stimuli, making them ideal for sensory functions (Sun et al., 2025).

S-type neurons (synaptic-type) are characterized by brief action potentials that lack slow afterhyperpolarization potentials and instead display fast excitatory postsynaptic potentials (EPSPs) (Fung et al., 2020) (Zhao et al., 2023). These neurons are relatively depolarized at rest and receive fast synaptic input, enabling rapid signal transmission and integration (Savulescu-Fiedler et al., 2025).

Morpho-electrophysiological correlations show strong alignment between electrical and structural properties. AH-type neurons typically display Dogiel Type II morphology with large, smooth cell bodies and multiple long processes, while S-type neurons correspond to Dogiel Type I morphology with flattened, elongated cell bodies and short dendrites (Kuwahara et al., 2020) (Dharshika et al., 2022) (Sayegh et al., 2013).

Functional correspondence reveals that AH-type neurons are predominantly intrinsic primary afferent neurons (IPANs) that serve as chemosensitive and mechanosensitive sensory elements of the ENS (Kuwahara et al., 2020) (Dharshika et al., 2022) (Zhao et al., 2023). In contrast, S-type neurons function as motor neurons, interneurons, and secretomotor neurons that mediate rapid signal transmission and control effector responses (Ren et al., 2008) (Dharshika et al., 2022) (Savulescu-Fiedler et al., 2025).

Modern molecular studies have validated these electrophysiological distinctions at the genetic level. Neurons expressing molecular markers characteristic of IPANs, such as Calcb and Nmu, consistently demonstrate AH-type electrophysiology along with the associated morphological features (Gomez-Frittelli et al., 2025) (Gomez-Frittelli et al., 2024). This integrated classification system continues to serve as a cornerstone for understanding enteric neural organization and has been conserved across multiple species studied, including guinea pigs, mice, and humans (Dharshika et al., 2022).

Evidence

(Fung et al., 2020)

Functional circuits and signal processing in the enteric nervous system

> *"Dogiel published illustrations of three morphological subtypes of neurons in the ENS, defined as Dogiel Types I, II, and III [14]. It was not until 1974 that Hirst and colleagues then reported on their electrophysiological properties and classified them into AH-and S-type neurons (Hirst et al., 1974). AH neurons are typified by their larger action potentials (APs) with an inflection on the falling phase and long afterhyperpolarizing potential (AHP; > 2 s) that follows. By comparison, S neurons are characterized by their brief APs that lack slow AHPs, and they display fast excitatory postsynaptic potentials (EPSPs)"*

> *"Costa, Furness, and colleagues conducted an extensive series of immunohistochemical and electron microscopy studies to group enteric neurons based on their distinct neurochemistry, projection patterns, and ultrastructure (Costa et al., 1996)(Costa et al., 1983)(Costa et al., 1981)(Costa et al., 1992)(Furness et al., 2004)(Furness et al., 1982)(Furness et al., 2004)[23][24][25][26][27][28][29]"*

> *"immunostaining for the expression of primary transmitters or their synthesizing enzymes, and cytoskeletal and calcium-binding proteins"*

> *"studies using various single-cell RNA-sequencing methods optimized for the ENS are now emerging, allowing for extensive gene expression profiling of functional molecular constituents of enteric neurons such as ion channels, transmitters and their synthesizing enzymes, as well as receptors [35,36]. This methodology based on statistical clustering algorithms subdivided myenteric neurons into nine molecularly defined subpopulations. By comparison, ten different classes of myenteric neurons were identified from integrated anatomical, neurochemical, functional, and pharmacological analysis (Furness et al., 2014)"*

> *"Intrinsic sensory neurons (or IPANs) are typically polymodal, possessing chemosensory and mechanosensory properties, and have Dogiel Type II morphology [40]. Furthermore, they are associated with AH-type characteristics where their AHP enables them to set a level of excitability through somatic gating of activity within the network [41]"*

> *"singlecell sequencing data provides evidence for several subclasses [35]36]. One clear division is whether they are situated in the submucosal or myenteric plexus. Only the former subset are sensitive to subtle mechanical stimulation of the mucosa"*

(Sayegh et al., 2013)

Back to Basics: Regulation of the Gastrointestinal Functions

> *"four main classification methods have been utilized. These methods depend on the [1] morphology (different shapes) of enteric neurons, [2] the types of neurotransmitters or peptides that they may contain (also known as their chemical coding), [3] the electrical properties of the enteric neurons or electrophysiology, and [4] the function (e.g., sensory, motor, inhibitory and excitatory) of the enteric neurons). On the basis of morphology the Russian histologist Alexander Dogiel described three types of enteric neurons in 1896 and 1899. These types are: (A) Dogiel type I, (B) Dogiel type II and (C) Dogiel type III neurons. Dogiel type I neurons have small, irregular cell bodies with multiple short dendrites. Dogiel type II neurons have large, oval-shaped cell bodies with one or two long dendrites. Dogiel type III neurons have large cell bodies with different shapes and multiple dendrites. Electrophysiologically, there*

are two types of enteric neurons: (I) S-type (S stands for synaptic) which evokes a fast (e.g., millisecond) action potential and (II) AH-neurons (AH stands for after hyperpolarization) which evoke longer lasting action potential (e.g., seconds) compared with the S-type"

"some enteric neurons contain the neurotransmitter acetylcholine. Such neurons are called cholinergic neurons and are generally stimulatory to gut activities. Other enteric neurons contain adrenaline (also known as epinephrine). These neurons are called adrenergic neurons and are generally inhibitory to gut activities"

"the enteric neurons can be classified on the basis of their function. The basic functions of enteric neurons are excitatory, inhibitory, sensory or motor. Excitatory neurons cause an increase in secretion if they innervate a gland or cause muscle contraction if they innervate a muscle. Inhibitory neurons cause a decrease in secretion or muscle relaxation. Sensory neurons detect luminal pH and pressure or temperature in the gut wall, while motor neurons innervate muscles and sphincters and cause contraction or relaxation. In general, Dogiel type I neurons and S-type enteric neurons are considered motor neurons, whereas Dogiel type II and AH-neurons are considered sensory neurons. The excitatory enteric neurons of the gut, which act in the orad (toward the mouth) direction, contain acetylcholine (Ach) and substance P (sub P), while the inhibitory enteric neurons of the gut, which act in the aborad (toward the anus) direction, contain Vasoactive Intestinal Peptide (VIP) and Nitric Oxide (NO)"

"myenteric and submucosal plexuses"

(Ren et al., 2008)

Purinergic receptors and synaptic transmission in enteric neurons

"There are enteric neurons that are sensitive to chemical or mechanical cues from the gastrointestinal tract with cell bodies in the myenteric and submucous plexes. They are properly referred to as intrinsic primary afferent neurons (IPANs) and are separate from the extrinsic primary afferent neurons whose cell bodies are in dorsal root or nodose ganglia (Furness et al., 1998). In this review we will simply refer to them as sensory neurons. Many enteric sensory neurons have "AH"-type electrophysiological characteristics which are defined as having an action potential afterhyperpolarization lasting >1 s (Hirst et al., 1974)(Bornstein et al., 1994)(Mao et al., 2006). Most sensory neurons have smooth cell bodies and are multipolar with projections that run to the mucosal epithelium, between the myenteric plexus and the submucosal plexus, and to their own and nearby ganglia (Pompolo et al., 1988). They form synapses with all other neuronal types"

"Enteric interneurons are primarily found in the myenteric plexus where there is generally one subtype of interneuron (two types in proximal colon) with an ascending projection and at least three subtypes of interneuron with a descending projection. There are also a small number of intestinofugal neurons that send a projection to prevertebral ganglia (Furness et al., 2002). All interneurons have a single axon, but there is a mix of dendritic morphology. Interneurons are characterized electrophysiologically as "S" neurons"

"In the myenteric plexus, ATP is utilized as a transmitter or co-transmitter by at least one type of descending interneuron during local reflexes (Bornstein, 2008)[17]. These descending interneurons are immunoreactive for nitric oxide synthase (NOS) and vasoactive intestinal polypeptide (VIP) and have Dogiel type I dendrites"

"Enteric motor neurons are found in the myenteric and submucous plexes. They have a single axon and short dendrites with Dogiel type I morphology. In the myenteric plexus, there are four functional classes of motor neuron to the smooth muscle layers of the intestine (Furness et al., 2002)(Brookes, 2001): inhibitory or excitatory motor neurons to either the circular or longitudinal smooth muscle layers. All motor neurons have S-type properties"

"In the myenteric plexus, inhibitory motor neurons to the circular muscle are immunoreactive for NOS and VIP and use ATP as a coneurotransmitter (Furness et al., 2002)(Costa et al., 1996)(Crist et al., 1992). In the submucous plexus, there are two types of excitatory secretomotor neurons going to the epithelium and at least one type of vasodilator neuron to the submucous arterioles. Noncholinergic secretomotor/ vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP. These neurons account for ~45% of submucosal neurons, and they may utilize ATP as a coneurotransmitter (Monro et al., 2004)(Hu et al., 2003)."

(Zhao et al., 2023)

The influence of Nav1.9 channels on intestinal hyperpathia and dysmotility

"Neurons in the ENS are mainly divided into Dogiel type I and Dogiel type II neurons in morphology. The cell bodies of Dogiel type I neurons project multiple short dendrites and a single elongated axon. Dogiel type II neurons have smooth cell bodies and project multiple axons, but no dendrites or only a few dendrites [70]. In addition, ENS neurons are divided into sensory neurons, interneurons and motor neurons in terms of functional characteristics [71], and AH-type neurons and S-type neurons in terms of electrophysiological characteristics [72]. AH-type neurons are characterized by the larger action potentials with an infection on the falling phase and long afterhyperpolarizing potential (AHP;>2 s) that follows. S-type neurons are typified by the brief action potentials without slow AHPs, and they present fast excitatory postsynaptic potentials (EPSPs) [68]. The intrinsic sensory neurons/ intrinsic primary afferent neurons (IPANs) of the ENS generally have the characteristics of Dogiel II neurons in morphology, and belong to AH neurons in electrophysiological characteristics [67,73,74]. The ENS comprises the myenteric plexus and the submucosal plexus. The myenteric plexus is mainly responsible for the regulation of intestinal movements. The submucosal plexus is involved in the secretion of water and electrolytes, as well as the neural regulation of intestinal blood flow [68]."

(Kuwahara et al., 2020)

[Microbiota-gut-brain axis: enteroendocrine cells and the enteric nervous system form an interface between the microbiota and the central nervous system.]

> *"Myenteric and submucosal neurons are composed of discrete populations of neurons that can be classified based on their function and morphology. These include intrinsic primary afferent neurons (IPANs) that allow them to regulate GI motility and secretion without CNS input, motor neurons (muscle, secretomotor and vasodilator neurons) and interneurons"*

> *"Neurons in the ENS are also divided into two subtypes based on their electrophysiological properties, which correlate with their morphologies (Furness, 2012). AH (after hyperpolarizing) neurons have multiple long processes and a large oval soma (AH/ Dogiel Type II neurons) and S (synaptic) neurons (Brookes et al., 1995)(Clerc et al., 1997)(Nurgali et al., 2004)). Morphology of S neuron is flattened, slightly elongated with stellate or angular forms. The characteristic belongs to Dogiel Type I neurons. S neurons function as muscular motors, secretomotors and interneurons. On the other hand, AH neurons are chemo-and mechanosensitive IPANs."*

(Sun et al., 2025)

[Gut sensory neurons as regulators of neuro-immune-microbial interactions: from molecular mechanisms to precision therapy for IBD/IBS]

> *"Morphologically classified as Dogiel type II multipolar neurons, their somata are predominantly localized within the myenteric plexus, with axonal terminals extensively projecting to the mucosal layer and adjacent plexuses, forming a dense sensory network (Spencer et al., 2020) (Fig. 1). Upon mechanical or chemical stimulation, IPANs generate action potentials through depolarization, releasing neurosignaling molecules such as tachykinins and CGRP"*

> *"Characterized as AH-type neurons, IPANs display prominent afterhyperpolarization (AHP) potentials and low-threshold action potentials, allowing sustained responsiveness to subthreshold stimuli [11]."*

(Savulescu-Fiedler et al., 2025)

[Rewiring the Brain Through the Gut: Insights into Microbiota–Nervous System Interactions]

> *"The ENS is structured with two main plexuses represented by the submucous plexus (Meissner) and the myenteric plexus (Auerbach)"*

> *"Both plexuses consist of three neuron types: the enteric intrinsic primary afferent neurons (IPANs), the motor neurons or efferent neurons, and the association neurons"*

> *"IPANs represent the most vulnerable ENS neuronal population, given the fact that they extend to the lamina propria of the intestine (Benarroch, 2007)(Furness et al., 2004)"*

> *"The enteric peristalsis is controlled by the motor neurons arising from the myenteric plexus, while absorption and secretion are regulated by the neurons emerging from the submucosal plexus (Furness et al., 2014)"*

> *"The main classes of enteric neurons are represented by neurons that receive fast synaptic input, known as "S" neurons, and neurons with prolonged post-hyperpolarization, referred to as "AH" neurons (Natale et al., 2021). IPANs belong to "AH" neurons, whereas motor and interneurons are "S" neurons, which are relatively depolarized at rest [69]."*

(Dharshika et al., 2022)

[Enteric Neuromics: How High-Throughput “Omics” Deepens Our Understanding of Enteric Nervous System Genetic Architecture]

> *"Type I neurons have flat, elongate, and irregular cell bodies with a single axon and numerous short dendrites, whereas type II neurons have smoother and larger cell bodies with multiple long axons. Type I neurons project and communicate with adjacent ganglia in the plexus and musculature, whereas type II neurons communicate with neurons throughout the gut wall, within and between ganglia, and the mucosa"*

> *"are also classified by electrophysiological properties, which have been mainly characterized in guinea pigs and mice. Two main types of enteric neurons are categorized as having either synaptic-or AH (after hyperpolarization)-type electrophysiological properties that differ on the basis of action potential speed and magnitude, the length of AH potentials, and tetrodotoxin sensitivity. Synaptic-type neurons typically display Dogiel type I morphology and include interneurons and motor neurons, whereas AH-type neurons typically display Dogiel type II morphology and are considered sensory neurons"*

> *"Defining the neurochemical coding of enteric neurons was a significant advancement in identifying neuronal subtypes and understanding how enteric neurons communicate with one another and target cells. Enteric neurochemical coding has been defined by multiple approaches including immunohistochemistry in combination with retrograde tracing, electrophysiology, and pharmacology. Integrating these biomolecular data with morphologic and electrophysiological properties is the basis for current definitions of enteric neuron subtypes, which include motor neurons, interneurons, and sensory neurons. Although these definitions are based largely on studies in guinea pigs, many of the core features of enteric neuron subtypes are conserved between mice and*

humans. Excitatory and inhibitory motor neurons reside in the myenteric plexus and innervate the circular and longitudinal muscle of the intestine."

(Gomez-Frittelli et al., 2025)

[Synaptic cell adhesion molecule Cdh6 identifies a class of sensory neurons with novel functions in colonic motility]

> *"Neuronal subtypes in the ENS have been distinguished by their electrophysiological properties, morphology, and expression of characteristic markers, notably neurotransmitters and neuropeptides"*

> *"Cdh6+ neurons demonstrate all other distinguishing classifications of enteric sensory neurons including marker expression of Calcb and Nmu, Dogiel type II morphology and AH-type electrophysiology and IH current"*

> *"These neuronal subtypes have begun to be distinguished morphologically, electrophysiologically, and by marker expression, classically especially of neurotransmitters (Nurgali et al., 2004)(Qu et al., 2008)"*

> *"Cdh6 neurons express IPAN markers Calcb and Nmu. Sparse labeling of individual IPANs reveals they project mainly circumferentially and branch extensively in myenteric ganglia."*

(Gomez-Frittelli et al., 2024)

[Synaptic cell adhesion molecule Cdh6 identifies a class of sensory neurons with novel functions in colonic motility]

> *"Neuronal subtypes in the ENS have been distinguished by their electrophysiological properties, morphology, and expression of characteristic markers, notably neurotransmitters and neuropeptides"*

> *"Cdh6+ neurons demonstrate all other distinguishing classifications of enteric sensory neurons including marker expression of Calcb and Nmu, Dogiel type II morphology and AH-type electrophysiology and IH current."*

Neurotransmitter and Molecular Markers

Enteric neurons utilize over 30 neurotransmitters and neuropeptides in complex combinations, with acetylcholine and nitric oxide being the most abundant, while modern single-cell sequencing has revealed distinct molecular signatures that define specific neuronal subtypes based on gene expression patterns. (20 sources)

Cholinergic neurotransmitters: Acetylcholine (ACh) represents the most prevalent neurotransmitter in the ENS, with cholinergic neurons comprising at least 70% of myenteric neurons (Pradeloux et al., 2025) (Li et al., 2003). Choline acetyltransferase (ChAT), the enzyme responsible for acetylcholine synthesis, serves as the primary marker for excitatory motor neurons (Jiang et al., 2017) (Petto et al., 2015). In the submucosal plexus, approximately 42% of inner submucosal plexus neurons and 32% of outer submucosal plexus neurons express ChAT (Makowska et al., 2019) (Petto et al., 2015).

Nitrergic neurotransmitters: Nitric oxide (NO) and nitric oxide synthase (NOS) represent the primary inhibitory neurotransmitters in the ENS (Magalhaes et al., 2021) (Gallego et al., 2016). Inhibitory motor neurons to the circular muscle are characterized by NOS and vasoactive intestinal polypeptide (VIP) immunoreactivity and use ATP as a co-neurotransmitter (Ren et al., 2008). In the outer submucosal plexus, 31% of neurons express nNOS, with most being exclusively nitrergic and 10% co-localizing with VIP (Makowska et al., 2019) (Petto et al., 2015).

Purinergic neurotransmitters: ATP serves as both a primary neurotransmitter and co-neurotransmitter, particularly important for inhibitory neuromuscular transmission alongside nitric oxide (Bornstein, 2008). ATP is utilized by at least one type of descending interneuron during local reflexes and acts through P2X and P2Y receptors to regulate intestinal motility (Ren et al., 2008) (Bornstein, 2008).

Peptidergic neurotransmitters: The ENS contains numerous neuropeptides including substance P, vasoactive intestinal peptide (VIP), somatostatin, neuropeptide Y (NPY), calcitonin gene-related peptide (CGRP), and others (Magalhaes et al., 2021) (Shen et al., 2024). Substance P is commonly co-expressed with acetylcholine, with approximately 66% of ChAT-positive neurons in the inner submucosal plexus co-localizing substance P (Makowska et al., 2019) (Petto et al., 2015).

Monoaminergic neurotransmitters: Serotonin serves dual roles within the ENS as both a neurotransmitter and paracrine signaling molecule (Wang et al., 2025). Dopaminergic neurons represent a small but functionally significant subset, accounting for approximately 10-13% of enteric neurons in mice and up to 20% in humans, with higher concentrations in the upper gastrointestinal tract (Pradeloux et al., 2025) (Li et al., 2003).

IPAN markers: Intrinsic primary afferent neurons are characterized by substance P, acetylcholine, calcitonin gene-related peptide (CGRP), and serotonin (Magalhaes et al., 2021). Single-cell sequencing studies have revealed distinct IPAN subtypes expressing different combinations of established markers including Calca/Calcb/Nfem/Calb1/Calb2, with clear separation based on selective expression of neuromedin U (Nmu), urocortin-3/cholecystokinin (Ucn-3/Cck), or neurexophilin-2 (Nxph2) (Morarach et al., 2020).

Motor neuron markers: Excitatory motor neurons are primarily identified by ChAT expression and co-expression of Tac1, while inhibitory motor neurons express Nos1, Gal, Vip, and Npy (Morarach et al., 2020). Traditional classification distinguishes neurons based on nitric oxide synthase 1 (NOS1) for inhibitory motor neurons and acetylcholine for excitatory motor neurons, as well as CALCA/B for IPANs (Majd et al., 2024).

Secretomotor/vasodilator markers: Non-cholinergic secretomotor and vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP, accounting for approximately 45% of submucosal neurons (Ren et al., 2008).

Modern single-cell RNA sequencing has revealed 12 distinct enteric neuron classes within the myenteric plexus, with some classes defined by selective expression of single genes including somatostatin (Sst), neuromedin U (Nmu), cholecystokinin (Cck), urocortin 3 (Ucn3), neuronal differentiation 6 (Neurod6), and neurexophilin-2 (Nxph2) (Morarach et al., 2020). Large-scale studies have profiled over 25,000 mouse neurons and 1,445 human neurons, revealing both conserved and species-specific transcriptional programs (Brehmer, 2021) (Drokhlyansky et al., 2020).

Broad molecular classifications: Enteric neurons can be grouped into two main categories based on primary neurotransmitters - NOS1-expressing neurons and ChAT-expressing neurons, with further subdivision based on gene expression patterns that correspond to classical functional categories (Jakob et al., 2021) (Zeisel et al., 2018).

Novel molecular markers: Recent studies have identified novel populations, including dopaminergic neurons characterized by co-expression of ChAT/DAT and molecular markers including Grp, Calcb, and Sst (Recinto et al., 2023). Additional diversity includes neurons expressing calcium-binding proteins like calretinin and calbindin, which show regional and developmental variation (Parathan et al., 2020).

The molecular complexity of the ENS continues to expand with technological advances, revealing that most neurons contain several neurotransmitters in distinctive co-localization patterns that identify functionally distinct neuronal sets (Cirillo et al., 2011).

Paper	Focus Of Neurotransmitter Or Molecular Markers	Methodology For Marker Identification	Ens Region Or Plexus Studied	Species Of Study Subject	Context Of Neurotransmitter Role	Classification Or Diversity Focus

Paper	Focus Of Neurotransmitter Or Molecular Markers	Methodology For Marker Identification	Ens Region Or Plexus Studied	Species Of Study Subject	Context Of Neurotransmitter Role	Classification Or Diversity Focus
Enteric Dopaminergic Neurons: Definition, Developmental Lineage, and Effects of Extrinsic Denervation	Dopamine (DA), tyrosine hydroxylase (TH), DA transporter (DAT)	immunocytochemistry, surgical ablation, extrinsic denervation	Submucosal and myenteric plexus, small intestine, stomach, colon	murine, guinea pig, mouse, rat	Developmental lineage of intrinsic enteric dopaminergic neurons.	classifies enteric dopaminergic neurons based on molecular markers
Visualizing the enteric nervous system using genetically engineered double reporter mice: Comparison with immunofluorescence	nNOS and ChAT	genetically engineered reporter mice, immunostaining	Esophagus, small intestine, and colon	mouse	control of gastrointestinal motility, secretion and absorption functions	Visualizing nNOS and ChAT positive neurons
Architecture and Chemical Coding of the Inner and Outer Submucous Plexus in the Colon of Piglets	Choline acetyltransferase (ChAT), substance P (SP), somatostatin (SOM), neuropeptide Y (NPY), vasoactive intestinal peptide (VIP), and neuronal nitric oxide synthase (nNOS).	Immunohistochemical staining was performed on wholemount preparations and cryostat sections using specific antibodies.	Inner and outer submucous plexuses in the porcine colon.	The study used porcine (pig) colon tissue.	Regulation of epithelial and circular muscle functions in the porcine colon.	Elucidates the diversity and classifies enteric neurons based on their neurotransmitter content and co-localization in the ISP and OSP.

Paper	Focus Of Neurotransmitter Or Molecular Markers	Methodology For Marker Identification	Ens Region Or Plexus Studied	Species Of Study Subject	Context Of Neurotransmitter Role	Classification Or Diversity Focus
Age and Sex-Dependent Differences in the Neurochemical Characterization of Calcitonin Gene-Related Peptide-Like Immunoreactive (CGRP-LI) Nervous Structures in the Porcine Descending Colon	Calcitonin gene-related peptide (CGRP) is the primary neurotransmitter/ molecular marker focused on.	Double-labeling immunofluorescence, microscopy, and quantification were used to identify and characterize markers.	Descending colon, including myenteric, outer submucous, and inner submucous plexuses.	The study used young and adult female and castrated male pigs.	The role of CGRP is examined in the context of age and sex-dependent differences during maturation in the porcine descending colon.	The paper elucidates the diversity of CGRP-positive neuronal types based on their neurotransmitter content and co-localization.
Diversification of molecularly defined myenteric neuron classes revealed by single cell RNA-sequencing	Histochemical markers for motor, sensory, and interneurons	Single-cell RNA sequencing, transcriptome analysis, imaging	myenteric plexus of the mouse small intestine	Mouse	intestinal motility, ENS development	Molecular taxonomy of 12 enteric neuron classes using scRNA-seq
Classification of human enteric neurons	neurochemical codes from multiple immunohistochemistry	Multiple immunohistochemistry combined with structural investigations	human gastrointestinal neuropathology	Human	human gastrointestinal neuropathology	classification based on morphology and neurochemical codes

Evidence

(Pradeloux et al., 2025)

[Impact of aging on the central and enteric nervous system in a Parkinson's disease mouse model]

"Functionally, enteric neurons are typically classified into five major categories based on their neurotransmitter profiles:

(1) excitatory motor neurons (cholinergic), (2) inhibitory motor neurons (nitrergic), (3) sensory neurons, (4) excitatory and inhibitory interneurons, and (5) secretomotor/vasodilator neurons (Natale et al., 2021). The cholinergic neurons make up the majority, at least 70% of myenteric neurons (Anlauf et al., 2003)"

"Those neurons, constituting less than 1% of all enteric cells, are organized into ganglionic plexuses-primarily the myenteric (MyP) and submucosal (SMP) plexuses-or into non-ganglionic nerve fiber bundles (Furness et al., 1980)(Brudek, 2019)"

"DAergic neurons represent a relatively small but functionally significant subset of the ENS. In mice, they account for approximately 10-13% of enteric neurons, while in humans, this proportion can reach up to 20%, with a distribution that follows an oral-to-aboral gradient (Anlauf et al., 2003)(Li et al., 2003). These neurons are particularly enriched in the upper gastrointestinal tract, where they constitute 14-20% of the neuronal population, but their prevalence drops to 1-6% in the distal small intestine. DAergic enteric neurons express key dopaminergic markers, including tyrosine hydroxylase (TH), dopamine transporter (DAT), and all five dopamine receptor subtypes (D1-D5)"

"Expression of the dopaminergic phenotype in the mouse ENS is variable depending on the region of the gut, TH transcripts being more prominent in the stomach and ileum compared to the colon (Chalazonitis et al., 2022)."

(Li et al., 2003)

Enteric Dopaminergic Neurons: Definition, Developmental Lineage, and Effects of Extrinsic Denervation

"The existence of enteric dopaminergic neurons has been suspected; however, the innervation of the gut by sympathetic nerves, in which dopamine (DA) is the norepinephrine precursor, complicates analyses of enteric DA. We now report that transcripts encoding tyrosine hydroxylase (TH) and the DA transporter (DAT) are present in the murine bowel (small intestine > stomach or colon; proximal colon > distal colon). Because sympathetic neurons are extrinsic, transcripts encoding TH and DAT in the bowel are probably derived from intrinsic neurons. TH protein was demonstrated immunocytochemically in neuronal perikarya (submucosal >> myenteric plexus; small intestine > stomach or colon). TH, DA, and DAT immunoreactivities were coincident in subsets of neurons (submucosal > myenteric) in guinea pig and mouse intestines in situ and in cultured guinea pig enteric ganglia. Surgical ablation of sympathetic nerves by extrinsic denervation of loops of the bowel did not affect DAT immunoreactivity but actually increased numbers of TH-immunoreactive neurons, expression of mRNA encoding TH and DAT, and enteric DOPAC (the specific dopamine metabolite). The fetal gut contains transiently catecholaminergic (TC) cells. TC cells are the proliferating crest-derived precursors of mature neurons that are not catecholaminergic and, thus, disappear after embryonic day (E) 14 (mouse) or E15 (rat). TC cells appear early in ontogeny, and their development/survival is dependent on mash-1 gene expression. In contrast, the intrinsic TH-expressing neurons of the murine bowel appear late (perinatally) and are mash-1 independent. We conclude that the enteric nervous system contains intrinsic dopaminergic neurons that arise from a mash-1-independent lineage of noncatecholaminergic precursors."

(Jiang et al., 2017)

Visualizing the enteric nervous system using genetically engineered double reporter mice: Comparison with immunofluorescence

"Each ganglion is a collection of many different types of neurons that can be classified based on the, 1) morphological appearance, 2) electrophysiological properties and, 3) chemical or neurotransmitter content (Furness et al., 2014)(Costa et al., 2000)(Carbone et al., 2014). The myenteric plexus resides between the circular and longitudinal muscle layers and is mostly responsible for the control of gastrointestinal motility. The majority of the myenteric plexus neurons can be divided into excitatory and inhibitory, which cause contraction and relaxation of the longitudinal and circular muscle layers. Acetylcholine (Ach) and substance P are the major neurotransmitters of excitatory motor neurons. On the other hand, nitric oxide (NO) and vasoactive intestinal peptide (VIP) are the major inhibitory neurotransmitters of inhibitory motor neurons"

"The latter is organized into two major plexuses, myenteric/ Auerbach and submucosal/Meissner, and several minor plexuses"

"The works of Brookes [4][5][6] in guinea pig, Sang in mice [7] and Wattchow in humans [8][9][10] show that majority of the myenteric neurons of small and large intestine contain either choline acetyl transferase (ChAT), the enzyme responsible for the synthesis of acetylcholine, or nitric oxide synthase (NOS), the enzyme responsible for the synthesis of nitric oxide."

(Petto et al., 2015)

Architecture and Chemical Coding of the Inner and Outer Submucous Plexus in the Colon of Piglets

"In the porcine colon, the submucous plexus is divided into an inner submucous plexus (ISP) on the epithelial side and an outer submucous plexus (OSP) on the circular muscle side. Although both plexuses are probably involved in the regulation of epithelial functions, they might differ in function and neurochemical coding according to their localization. Therefore, we examined expression and co-localization of different neurotransmitters and neuronal markers in both plexuses as well as in neuronal fibres. Immunohistochemical staining was performed on wholemount preparations of ISP and OSP and on cryostat sections. Antibodies against choline acetyltransferase (ChAT), substance P (SP), somatostatin (SOM), neuropeptide Y (NPY), vasoactive intestinal peptide (VIP), neuronal nitric oxide synthase (nNOS) and the pan-neuronal markers Hu C/D and neuron specific enolase (NSE) were used. The ISP contained 1,380 ± 131 ganglia per cm2 and 122 ± 12 neurons per ganglion. In contrast, the OSP showed a wider meshwork (215 ± 33 ganglia per cm2) and smaller ganglia (57 ± 3 neurons per ganglion). In the ISP, 42% of all neurons expressed

ChAT. About 66% of ChAT-positive neurons co-localized SP. A small number of ISP neurons expressed SOM. Chemical coding in the OSP was more complex. Besides the ChAT/±SP subpopulation (32% of all neurons), a nNOS-immunoreactive population (31%) was detected. Most nitrergic neurons were only immunoreactive for nNOS; 10% co-localized with VIP. A small subpopulation of OSP neurons was immunoreactive for ChAT/nNOS/±VIP. All types of neurotransmitters found in the ISP or OSP were also detected in neuronal fibres within the mucosa. We suppose that the cholinergic population in the ISP is involved in the control of epithelial functions. Regarding neurochemical coding, the OSP shares some similarities with the myenteric plexus. Because of its location and neurochemical characteristics, the OSP may be involved in controlling both the mucosa and circular muscle."

(Makowska et al., 2019)

Age and Sex-Dependent Differences in the Neurochemical Characterization of Calcitonin Gene-Related Peptide-Like Immunoreactive (CGRP-LI) Nervous Structures in the Porcine Descending Colon

> *"The enteric plexuses consist of millions of neurons, which play various roles and express a range of neurotransmitters and/or neuromodulators (Furness et al., 2014)"*

> *"Depending on the animal species and the fragment of the GI tract, the ENS structure shows some differences. In the esophagus and stomach, the enteric neurons of large mammals (e.g., pigs) are grouped into two ganglionated plexuses: the myenteric plexus (MP)-located between longitudinal and circular muscle layers and the submucous plexus placed near the lamina propria of the mucosal layer. In the small and large intestines, the submucous plexus is additionally divided into the outer submucous plexus (OSP) and inner submucous plexus (ISP), which are placed in the submucosal layer: the OSP near the inner side of the circular muscle layer and the ISP near the lamina propria of the mucosa (Balemba et al., 1998)(Petto et al., 2015)(Balemba et al., 1999)"*

> *"enteric neurons are very diverse in terms of morphological, functional and electrophysiological properties, but the most important criterion of neuronal classification in the ENS is the neurochemical characterization of nerve cells (Balemba et al., 1998)(Petto et al., 2015). To date, several dozen neuronal active substances have been found in the enteric neurons. They may play various functions and are involved in all aspects of gastrointestinal physiology, such as motility, excretive activity, intestinal blood flow and ion transport [1]6]."*

(Magalhaes et al., 2021)

Enteric nervous system and inflammatory bowel diseases: Correlated impacts and therapeutic approaches through the P2X7 receptor

> *"As proposed by Aleksandr S. Dogiel in 1899, the morphological classification of enteric neurons can be based on their conformation and dendritic distribution. Dogiel described type I cells as flattened, slightly elongated, with an angled or star-shaped contour, and, as remarkable characteristics, as having only one axon and four to 20 Lamellar dendrites that frequently extend at a short distance from the cell body [31]"*

> *"Type II neurons have large round or oval cell bodies and eccentric nuclei (Pompolo et al., 1988), and the surface is grooved by bundles of neural fibers (Hendriks et al., 1990). The main characteristic of type II neurons is the presence of several axonal processes that are emitted either directly from the cell body (multipolar neuron) or from a single initial process that branches into short subsidiary axons (pseudounipolar neurons) (Furness, 2012)(Song et al., 1994)"*

> *"Additionally, enteric neurons can also be identified as intrinsic primary afferent neurons (IPANs), interneurons, and motor neurons (Furness, 2012), classified into at least 18 subtypes and using more than 30 neurotransmitters in their synapses [28]30]. Of these neurotransmitters, acetylcholine (ACh) and nitric oxide (NO) stand out as the most abundant (Brookes, 2001), as well as adenosine-5'-triphosphate (ATP) (Hansen, 2003), vasoactive intestinal polypeptide (VIP), and substance P (SP) (Gallego et al., 2016)"*

> *"IPANs (classified as Dogiel type II) are recognized for responding to chemical stimuli, mucosal deformation and GI muscle tension, translating these signals into a neural impulse that will trigger a local motor reflex (Costa et al., 2000). Altogether, IPANs represent approximately 14% and 30% of the neurons of the submucosal and myenteric plexuses, respectively"*

> *"The enteric neural circuit is arranged in two plexuses: the submucosal plexus, which in large mammals is present in two individualized levels (outer/inner) and is located in the outer connective tissue layer and the inner mucosal layer, and the myenteric plexus, located between the longitudinal and circular muscle layers (Bayliss et al., 1899)(Furness, 2012)5,28]"*

(Gallego et al., 2016)

Mechanisms responsible for neuromuscular relaxation in the gastrointestinal tract.

> *"The enteric nervous system (ENS) is responsible for the genesis of motor patterns ensuring an appropriate intestinal transit. Enteric neurons are classified into afferent, interneuron, and motoneuron types, with the latter two being further categorized as excitatory or inhibitory, which cause smooth muscle contraction or inhibition, respectively. Muscle relaxation mechanisms are key for the understanding of physiological processes such as sphincter relaxation, gastric accommodation, or descending peristaltic reflex. Nitric oxide (NO) and ATP or a related purine represent the primary inhibitory neurotransmitters. Nitrergic neurons synthesize NO through nNOS enzyme activity. NO diffuses across the cell membrane to bind its receptor, namely, guanylyl cyclase, and then activates a number of intracellular mechanisms that ultimately result in muscle relaxation. ATP acts as an inhibitory*

neurotransmitter together with NO, and the purinergic P2Y1 membrane receptor has been identified as a key item in order to understand how ATP may relax intestinal smooth muscle. Although, probably, no clinician doubts the significance of NO in the pathophysiology of digestive motility, the relevance of purinergic neurotransmission is apparently much lower, as ATP has not been associated with any specific motor dysfunction yet. The goal of this review is to discuss the function of both relaxation mechanisms in order to establish the physiological grounds of potential motor dysfunctions arising from impaired intestinal relaxation."

(Ren et al., 2008)

Purinergic receptors and synaptic transmission in enteric neurons

> *"There are enteric neurons that are sensitive to chemical or mechanical cues from the gastrointestinal tract with cell bodies in the myenteric and submucous plexes. They are properly referred to as intrinsic primary afferent neurons (IPANs) and are separate from the extrinsic primary afferent neurons whose cell bodies are in dorsal root or nodose ganglia (Furness et al., 1998). In this review we will simply refer to them as sensory neurons. Many enteric sensory neurons have "AH"-type electrophysiological characteristics which are defined as having an action potential afterhyperpolarization lasting >1 s (Hirst et al., 1974)(Bornstein et al., 1994)(Mao et al., 2006). Most sensory neurons have smooth cell bodies and are multipolar with projections that run to the mucosal epithelium, between the myenteric plexus and the submucosal plexus, and to their own and nearby ganglia (Pompolo et al., 1988). They form synapses with all other neuronal types"*

> *"Enteric interneurons are primarily found in the myenteric plexus where there is generally one subtype of interneuron (two types in proximal colon) with an ascending projection and at least three subtypes of interneuron with a descending projection. There are also a small number of intestinofugal neurons that send a projection to prevertebral ganglia (Furness et al., 2002). All interneurons have a single axon, but there is a mix of dendritic morphology. Interneurons are characterized electrophysiologically as "S" neurons"*

> *"In the myenteric plexus, ATP is utilized as a transmitter or co-transmitter by at least one type of descending interneuron during local reflexes (Bornstein, 2008)[17]. These descending interneurons are immunoreactive for nitric oxide synthase (NOS) and vasoactive intestinal polypeptide (VIP) and have Dogiel type I dendrites"*

> *"Enteric motor neurons are found in the myenteric and submucous plexes. They have a single axon and short dendrites with Dogiel type I morphology. In the myenteric plexus, there are four functional classes of motor neuron to the smooth muscle layers of the intestine (Furness et al., 2002)(Brookes, 2001): inhibitory or excitatory motor neurons to either the circular or longitudinal smooth muscle layers. All motor neurons have S-type properties"*

> *"In the myenteric plexus, inhibitory motor neurons to the circular muscle are immunoreactive for NOS and VIP and use ATP as a coneurotransmitter (Furness et al., 2002)(Costa et al., 1996)(Crist et al., 1992). In the submucous plexus, there are two types of excitatory secretomotor neurons going to the epithelium and at least one type of vasodilator neuron to the submucous arterioles. Noncholinergic secretomotor/ vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP. These neurons account for ~45% of submucosal neurons, and they may utilize ATP as a coneurotransmitter (Monro et al., 2004)(Hu et al., 2003)."*

(Bornstein, 2008)

Purinergic mechanisms in the control of gastrointestinal motility

> *"For many years, ATP and adenosine have been implicated in movement regulation of the gastrointestinal tract. They act through three major receptor subtypes: adenosine or P1 receptors, P2X receptors and P2Y receptors. Each of these major receptor types can be subdivided into several different classes and is widely distributed amongst various neurons, muscle types, glia and interstitial cells that regulate intestinal functions. Several key roles for the different receptors and their endogenous ligands have been identified in physiological and pharmacological studies. For example, adenosine acting at A1 receptors appears to inhibit intestinal motility in various pathological conditions. Similarly, ATP acting at P2Y receptors is an important component of inhibitory neuromuscular transmission, acting as a cotransmitter with nitric oxide. ATP acting at P2X and P2Y1 receptors is important for synaptic transmission in simple descending excitatory and inhibitory reflex pathways. Some P2Y receptor subtypes prefer uridine nucleotides over purine nucleotides. Thus, roles for UTP and UDP as enteric transmitters in place of ATP cannot be excluded. ATP also appears to be important for sensory transduction, especially in chemosensitive pathways that initiate local inhibitory reflexes. Despite this evidence, data are lacking about the roles of either adenosine or ATP in more complex motility patterns such as segmentation or the interdigestive migrating motor complex. Clarification of roles for purinergic transmission in these common, but understudied, motility patterns will depend on the use of subtype-specific antagonists that in some cases have not yet been developed."*

(Shen et al., 2024)

Systematic Insights into the Relationship between the Microbiota–Gut–Brain Axis and Stroke with the Focus on Tryptophan Metabolism

> *"The ENS comprises enteric neurons and enteric glia cells, forming two main plexuses, the submucous (or Meissner's) plexus between the circular muscle layer and the muscularis mucosa, innervating the muscle motor, and the myenteric (or Auerbach's) plexus between the longitudinal and circular muscle, responsible for secretory and absorptive control. Neurons of the ENS can be classified by function into intrinsic primary afferent neurons (IPANs), interneurons, motor neurons, and intestinofugal neurons (IFN), or by morphology, mainly into Dogiel type I and type II neurons (Benarroch, 2007)(Spencer et al., 2020). In general, primary*

transmitters in enteric neurons include acetylcholine (Ach), serotonin (5-HT), nitric oxide (NO), substance P, vasoactive intestinal peptide (VIP), and calcitonin gene-related peptide (CGRP). The myenteric neurons are mostly cholinergic or nitrergic, while submucosal neurons are mainly cholinergic neurons or VIP-expressing noncholinergic neurons (Benarroch, 2007)."

(Wang et al., 2025)

[Dietary Modulation of the Enteric Nervous System: From Molecular Mechanisms to Therapeutic Applications]

"Neurons within the ENS are categorized according to various characteristics, such as morphology, neurotransmitter expression, cell surface receptors, electrophysiological properties, connectivity patterns, their specific location within the intestinal wall, and their functional roles. Based on the morphological characteristics of ENS neurons, particularly the shape and length of their dendrites, Dogiel classified these neurons into types Dogiel I-III [28]. With the advent of advanced technologies, notably the application of single-cell RNA sequencing, a more intricate heterogeneity of neurons within the ENS has been uncovered. Through single-nucleus analysis of the human intestine, researchers have identified 436,202 nuclei and successfully isolated 1445 neurons [29]. The functionality of the ENS is dependent on the coordinated action of diverse neurotransmitters, which can be categorized according to their chemical properties and functions. Among cholinergic neurotransmitters, acetylcholine (ACh) is the most prevalent and plays a pivotal role in the ENS [22]"

"Monoaminergic neurotransmitters: Serotonin is a crucial monoamine neurotransmitter with dual roles within the ENS."

(Morarach et al., 2020)

[Diversification of molecularly defined myenteric neuron classes revealed by single cell RNA-sequencing]

"Meticulous work over the last decades has identified a framework of enteric neurons with sensory (Intrinsic Primary Afferent Neurons, IPANs), motor or interneuron characteristics"

"we define a molecular taxonomy of 12 enteric neuron classes within the myenteric plexus of the mouse small intestine using single-cell RNA sequencing. We present cell–cell communication features and histochemical markers for motor neurons, sensory neurons and interneurons"

"some classes were defined by the selective expression of single genes with likely importance for their neuronal activity. These included ENC5 (somatostatin; Sst), ENC6 (neuromedin U; Nmu), ENC7 (cholecystokinin; Cck and urocortin 3; Ucn3), ENC10 (neuronal differentiation 6; Neurod6) and ENC12 (neurexophilin-2; Nxph2). Other classes were distinguished by combinatorial gene codes"

"ENC1-4 displayed a partly shared gene expression, including Tac1, but differed in their expression of Calb2, Ndufa4l2, Gda, Penk and Fut9"

"ENC8-9 jointly expressed Nos1 and c1ql1, but Npy and Cox8b were significantly lower in ENC9. ENC11 highly expressed Npy, but lacked Nos1"

"ENC1-4 were assigned to excitatory motor neurons (Tac1/Calb2), while ENC8 and 9 matched inhibitory motor neurons (Nos1/Gal/Vip/Npy). ENC6, 7 and 12 were mapped to IPANs and expressed different combinations of previously defined markers (Calca/Calcb/Nfem/Calb1/Calb2). These plausible IPANs were clearly separable by their selective expression of Nmu, Ucn-3/Cck or Nxph2."

(Majd et al., 2024)

[A call for a unified and multimodal definition of cellular identity in the enteric nervous system]

"For decades, enteric neuron subtype identity has largely been defined based on morphological, immunohistochemical and functional features of neurons within the gut, including neuronal projection localization, neurochemistry, and electrophysiological properties. Investigations conducted on tissues isolated from model organisms, such as the guinea pig, mouse and rat 1 have significantly progressed our understanding of the neurochemical and functional complexity of enteric neuron subtypes and have identified functional units within the ENS circuitry. These include intrinsic primary afferent neurons (IPANs or sensory neurons, SNs), ascending and descending interneurons (IN), excitatory and inhibitory motor neurons (EMNs and IMNs), and secretomotor and/or vasodilator neurons (SVNs) 1 . Despite the fact that many known neurotransmitters and neuropeptides are present in the ENS, only a handful are consistently used as markers to identify major cell types in the ENS. Some examples include nitric oxide synthase (NOS1) for IMNs and acetylcholine for EMNs as well as CALCA/B for IPANs 2 . While it is impressive how large swaths of ENS function are coordinated by these neurotransmitters and neuropeptides, it also raises the question of whether yet-to-be-determined factors confer cell-type specific functional properties 3 . These features might include secreted peptides and proteins, electrophysiological features mediated by ion channels and transporters, metabolic profiles, and specialized capabilities for cell-cell interaction or synapse formation. To accurately understand the identity of individual cell types, it is also crucial to consider the contextual information in which they operate, including their position in the gut. Histochemical and functional assays have identified differences along the length of the GI tract, as well as between species, highlighting the need for more systematic efforts towards comprehensive mapping and profiling of the ENS 1 ."

(Brehmer, 2021)

Classification of human enteric neurons

"both animal and human enteric neurons are usually divided into only two morphological subpopulations, "Dogiel type II" neurons (with several long processes) and "Dogiel type I" neurons (with several short processes). This implies no more than the distinction of intrinsic primary afferent from all other enteric neurons."

(Drokhlyansky et al., 2020)

The Human and Mouse Enteric Nervous System at Single-Cell Resolution.

"The enteric nervous system (ENS) coordinates diverse functions in the intestine but has eluded comprehensive molecular characterization because of the rarity and diversity of cells. Here we develop two methods to profile the ENS of adult mice and humans at single-cell resolution: RAISIN RNA-seq for profiling intact nuclei with ribosome-bound mRNA and MIRACL-seq for label-free enrichment of rare cell types by droplet-based profiling. The 1,187,535 nuclei in our mouse atlas include 5,068 neurons from the ileum and colon, revealing extraordinary neuron diversity. We highlight circadian expression changes in enteric neurons, show that disease-related genes are dysregulated with aging, and identify differences between the ileum and proximal/distal colon. In humans, we profile 436,202 nuclei, recovering 1,445 neurons, and identify conserved and species-specific transcriptional programs and putative neuro-epithelial, neuro-stromal, and neuro-immune interactions. The human ENS expresses risk genes for neuropathic, inflammatory, and extra-intestinal diseases, suggesting neuronal contributions to disease."

(Jakob et al., 2021)

An Integrated View on Neuronal Subsets in the Peripheral Nervous System and Their Role in Immunoregulation

"The somas of enteric neurons mainly cluster in two anatomically distinct but strongly interconnected ganglionated plexuses (10). The outer, myenteric plexus (Auerbach Plexus) lies within the longitudinal and the circular muscle layer, and the inner, submucosal plexus (Meissner Plexus) is located below the muscle layers (Jakob et al., 2020)"

"The intrinsic micro-circuitry regulation of intestinal function consists of five neuronal classes with distinct functional specializations. Sensory neurons, also referred to as intrinsic primary afferent neurons (IPANs), detect chemical and physical alterations in the intestine and transmit the signal via interneurons to excitatory motor neurons, inhibitory motor neurons or secretomotor/vasodilator neurons"

"Single-cell sequencing of the ENS using fluorescenceactivated sorting of Wnt1-Cre; R26Tomato mice revealed 1105 enteric neurons (with ~3066 genes detected on average) and eventually found nine clusters of neurons in the muscular sheet of the intestine (Zeisel et al., 2018). Based on neuro-transmitters nitric oxide synthase 1 (NOS1) and choline acetyltransferase (ChAT), two main groups can be classified, NOS1 + neurons and ChAT + neurons. The authors assigned three distinct neuronal clusters of nitrergic neurons, which comprise inhibitory motor neurons and secretomotor/vasodilator neurons (Table 1). Further, 6 clusters of cholinergic neurons, which express ChAT and the solute carrier family 5 member 7 (Slc5a7, gene encodes for an ion transporter) could be distinguished (Zeisel et al., 2018). ChAT + neurons are further subdivided into excitatory motor neurons, sensory neurons and interneurons according to their gene expression (Table 1)"

"Using the pan-neuronal Baf53b-cre deleter mice crossed to R26R-tomato to enable sort-purification of enteric neurons from the myenteric plexus of the small intestine combined with singlecell RNA-sequencing, 4,892 high-quality enteric neurons have been analyzed and 12 classes of neurons could be identified based on their gene expression (13)."

(Zeisel et al., 2018)

Molecular Architecture of the Mouse Nervous System

"The mammalian nervous system executes complex behaviors controlled by specialised, precisely positioned and interacting cell types. Here, we used RNA sequencing of half a million single cells to create a detailed census of cell types in the mouse nervous system. We mapped cell types spatially and derived a hierarchical, data-driven taxonomy. Neurons were the most diverse, and were grouped by developmental anatomical units, and by the expression of neurotransmitters and neuropeptides. Neuronal diversity was driven by genes encoding cell identity, synaptic connectivity, neurotransmission and membrane conductance. We discovered several distinct, regionally restricted, astrocytes types, which obeyed developmental boundaries and correlated with the spatial distribution of key glutamate and glycine neurotransmitters. In contrast, oligodendrocytes showed a loss of regional identity, followed by a secondary diversification. The resource presented here lays a solid foundation for understanding the molecular architecture of the mammalian nervous system, and enables genetic manipulation of specific cell types."

(Recinto et al., 2023)

Characterizing enteric neurons in Dopamine Transporter (DAT)-Cre reporter mice reveals dopaminergic subtypes with dual-transmitter content

"We observed distinct subtypes of DAT-tdTomato neurons expressing co-transmitters and modulators across both plexuses; some of them likely co-releasing acetylcholine, while others were positive for a slew of canonical DA markers (TH, VMAT2 and GIRK2). Interestingly, we uncovered a seemingly novel population of DA neurons unique to the ENS which were ChAT/DAT-tdTomato-immunoreactive neurons and were characterised by the expression of Grp, Calcb and Sst."

(Parathan et al., 2020)

The enteric nervous system undergoes significant chemical and synaptic maturation during adolescence in mice.

> *"Adolescence is a critical period of development. It is very likely that there is significant maturation of the enteric nervous system (ENS) of the gut during this stage of life, especially since there are substantial changes in factors known to influence the ENS including diet and microbiota during this time, but this remains unknown. To examine maturation of the ENS during adolescence, we performed immunohistochemistry using advanced microscopy and analytical methods to compare enteric neurons and glia of the duodenum and colon of mice taken prior to weaning with those of young adult mice. We found significant changes in the architecture of both myenteric and submucosal plexuses and surprisingly found subsets of enteric cells that co-expressed the pan-neuronal marker, Hu, and either glial markers Sox10 or S100β, not both. About 70% and 35% of all Hu + neurons in the submucous plexus of the young adult duodenum and colon respectively also expressed S100β. The proportion of Hu+/Sox10 + cells in the duodenal myenteric plexus decreased, while the proportion of Hu+/S100β+ cells in the colonic submucosal plexus increased during adolescence. In the submucous plexus, there were significant increases in the proportions of vasoactive intestinal peptide+ and choline acetyltransferase + secretomotor neurons, of neurofilament M (NFM)+ neurons in the colon and of calretinin + neurons in the duodenum during adolescence. There were no age-dependent changes in the neurochemistry of various myenteric neuronal subtypes, including those immunoreactive for neuronal nitric oxide synthase (nNOS), Calbindin, Calretinin or NFM. There were significant increases in the somata sizes of Calretinin + submucosal and myenteric neurons, and nNOS + myenteric neurons, and these enteric neurons received significantly more synaptophysin + contacts onto their cell bodies during adolescence. This is the first study showing that enteric neurons and glia in the gut undergo significant changes in their anatomy and chemistry during adolescence. Notably changes in synaptic contacts within the enteric circuitry strongly suggest maturation in gastrointestinal function occurs during this time."*

(Cirillo et al., 2011)

Enteric Nervous System Abnormalities in Ulcerative Colitis

> *"Although up to eight morphologic forms of neurons have been identified in the ENS, there are two main types: type I neurons, that have many club-shaped processes and a single long process, and type II neurons, that are multipolar and have many long, smooth processes (Furness et al., 1980). Functionally, enteric neurons can be classified into primary afferent neurons, interneurons and motor neurons, synaptically linked to each other in microcircuits (Furness et al., 1980). Moreover, there is a general classification between neurons of the submucosal plexus and neurons of the myenteric plexus: while the first ones predominantly innervate the mucosa and regulate secretion, absorption and blood flow, the second ones are primarily involved in the control of intestinal motility (Brookes, 2001)"*

> *"The chemical neuro-mediators of the ENS were initially thought to be limited to neurotransmitters such as acetylcholine and serotonin, but, subsequently, purines, such as ATP, amino acids, such as -aminobutyric acid, and peptides, such as vasoactive intestinal polypeptide (VIP) and neuropeptide Y (NPY), were identified (Gershon et al., 1994). More recently, nitric oxide (NO) has emerged as an important neurotransmitter in the ENS (Bogers et al., 1994). Overall, more than 20 candidate neurotransmitters have now been identified in enteric neurons, and most neurons contain several of them (Gershon et al., 1994). Distinctive patterns of co-localization of mediators appear to identify sets of neurons that perform distinct functions (Costa & Brookes, 1994"*

Anatomical Specialization Across Plexuses

The ENS is organized into three anatomically distinct ganglionated plexuses with specialized functions: the myenteric plexus controls gut motility, the submucosal plexus regulates secretion and blood flow, and the subserosal plexus provides additional innervation in some species. Each plexus contains specific neuronal populations with distinct morphological, functional, and neurochemical characteristics tailored to their anatomical location and physiological roles. (14 sources)

The enteric nervous system displays remarkable anatomical specialization across its three main ganglionated plexuses, with each plexus containing distinct neuronal populations optimized for specific physiological functions (Bosi et al., 2020) (Xu et al., 2025).

The myenteric plexus (Auerbach's plexus) is positioned between the longitudinal and circular muscle layers and serves as the primary control center for gastrointestinal motility (Jiang et al., 2017) (Jakob et al., 2021). This plexus contains the majority of enteric neurons, with approximately 30% being IPANs that detect mechanical and chemical stimuli (Magalhaes et al., 2021). The myenteric plexus houses four functional classes of motor neurons: excitatory and inhibitory motor neurons to both the circular and longitudinal smooth muscle layers (Ren et al., 2008). Excitatory motor neurons are primarily cholinergic, expressing choline acetyltransferase (ChAT), while inhibitory motor neurons are characterized by nitric oxide synthase (NOS) and vasoactive intestinal polypeptide (VIP) immunoreactivity (Ren et al., 2008) (Jiang et al., 2017). Interneurons are predominantly located in the myenteric plexus, with one subtype of ascending interneuron and at least three subtypes of descending interneurons (Ren et al., 2008).

The submucosal plexus (Meissner's plexus) is located in the submucosal layer and primarily regulates secretory activity, water and electrolyte transport, and local blood flow (Zhao et al., 2023) (Bubeck et al., 2023). In large mammals, this plexus is subdivided into two anatomically distinct layers: the outer submucosal plexus (OSP) near the circular muscle layer and the inner submucosal plexus (ISP) near the lamina propria (Makowska et al., 2019) (Nikolla et al., 2025). The inner submucosal plexus shows higher ganglia density (1,380 ± 131 ganglia per cm^2) with larger ganglia (122 ± 12 neurons per ganglion), while the outer submucosal plexus displays a wider meshwork (215 ± 33 ganglia per cm^2) with smaller ganglia (57 ± 3 neurons per ganglion) (Makowska et al., 2019). Neurochemically, the inner submucosal plexus contains predominantly cholinergic neurons (42% expressing ChAT), with about 66% of ChAT-positive neurons co-localizing substance P (Makowska et al., 2019). The outer submucosal plexus shows more complex chemical coding, containing both cholinergic (32%) and nitrergic (31%) populations (Makowska et al., 2019).

The subserosal plexus represents the third, less prominent ganglionated plexus located on the serosal surface, though its presence varies across species and gut regions (Bosi et al., 2020) (Xu et al., 2025).

Functional specialization reflects the anatomical positioning of each plexus. IPANs are present in both myenteric and submucosal plexuses, but only submucosal IPANs are sensitive to subtle mechanical stimulation of the mucosa (Ren et al., 2008). The submucosal plexus contains specialized secretomotor neurons that project to the epithelium and vasodilator neurons that innervate submucosal arterioles, with non-cholinergic secretomotor/vasodilator neurons expressing dynorphin, galanin, and VIP comprising approximately 45% of submucosal neurons (Ren et al., 2008). Motor control of the muscularis externa is primarily mediated by myenteric neurons, while epithelial secretion and absorption are controlled by submucosal neurons (Savulescu-Fiedler et al., 2025).

Regional variations in plexus organization occur along the gastrointestinal tract. In smaller mammals, myenteric neurons primarily regulate motility while submucosal neurons control secretion and vascular tone, but in larger mammals, some submucosal neurons also participate directly in motility reflexes (Hao et al., 2009). The esophagus and stomach contain simpler plexus arrangements compared to the small and large intestines, where the submucosal plexus shows clear anatomical subdivision into outer and inner layers (Makowska et al., 2019).

Single-cell transcriptomic studies have validated these anatomical specializations at the molecular level, revealing that plexus-specific gene expression patterns correspond to the distinct functional requirements of each anatomical compartment (Jakob et al., 2021) (Zeisel et al., 2018). This anatomical organization enables the ENS to maintain precise local control over diverse gastrointestinal functions through specialized neural circuits optimized for their specific anatomical niches (Llorente, 2024).

Evidence

(Bosi et al., 2020)

Tryptophan Metabolites Along the Microbiota-Gut-Brain Axis: An Interkingdom Communication System Influencing the Gut in Health and Disease

> *"The ENS is characterized by 3 major ganglionated plexuses, the submucosal, myenteric, and subserous plexus, separated by interconnecting fibre strands. nteric neurons are classified according to their morphology, neurochemical coding, projections to targets, and functional roles into 5 major types: intrinsic primary afferent neurons (IPAN), interneurons, excitatory and inhibitory motor neurons, and secretomotor neurons. In the submucosal and myenteric plexus, IPANs possess large-cell bodies and bidirectional axons projecting to the mucosa to respond to chemical and mechanical stimuli giving rise to secretomotor, vasodilator, and motor reflexes. This neuronal subtype is highly conserved in different animal species and can be identified by the presence of hyperpolarization-activated cation current and post-action potential and by the expression of different neurotransmitters such as substance P, acetylcholine, and calcitonin gene-related peptide and 5-HT."*

(Xu et al., 2025)

Tryptophan and Its Metabolite Serotonin Impact Metabolic and Mental Disorders via the Brain–Gut–Microbiome Axis: A Focus on

Sex Differences

"The ENS consists of three anatomical ganglionated plexuses of submucosal, myenteric, and subserous plexuses, interconnected by fibers (O'Mahony et al., 2015). Five major types of enteric neurons, based on their morphological, neurochemical, and functional characteristics, are intrinsic primary afferent neurons, interneurons, and excitatory, inhibitory, and secreto-motor neurons [38] (Furness, 2012). These neurons form a bidirectional network, with afferent neurons in the submucosal and myenteric plexuses responding to chemical and mechanical stimuli and regulating vasodilation and blood flow, motility, mucosal transport, secretion, and nutrient absorption (Furness et al., 2004)."

(Jiang et al., 2017)

Visualizing the enteric nervous system using genetically engineered double reporter mice: Comparison with immunofluorescence

"Each ganglion is a collection of many different types of neurons that can be classified based on the, 1) morphological appearance, 2) electrophysiological properties and, 3) chemical or neurotransmitter content (Furness et al., 2014)(Costa et al., 2000)(Carbone et al., 2014). The myenteric plexus resides between the circular and longitudinal muscle layers and is mostly responsible for the control of gastrointestinal motility. The majority of the myenteric plexus neurons can be divided into excitatory and inhibitory, which cause contraction and relaxation of the longitudinal and circular muscle layers. Acetylcholine (Ach) and substance P are the major neurotransmitters of excitatory motor neurons. On the other hand, nitric oxide (NO) and vasoactive intestinal peptide (VIP) are the major inhibitory neurotransmitters of inhibitory motor neurons"

"The latter is organized into two major plexuses, myenteric/ Auerbach and submucosal/Meissner, and several minor plexuses"

"The works of Brookes [4][5][6] in guinea pig, Sang in mice [7] and Wattchow in humans [8][9][10] show that majority of the myenteric neurons of small and large intestine contain either choline acetyl transferase (ChAT), the enzyme responsible for the synthesis of acetylcholine, or nitric oxide synthase (NOS), the enzyme responsible for the synthesis of nitric oxide."

(Jakob et al., 2021)

An Integrated View on Neuronal Subsets in the Peripheral Nervous System and Their Role in Immunoregulation

"The somas of enteric neurons mainly cluster in two anatomically distinct but strongly interconnected ganglionated plexuses (10). The outer, myenteric plexus (Auerbach Plexus) lies within the longitudinal and the circular muscle layer, and the inner, submucosal plexus (Meissner Plexus) is located below the muscle layers (Jakob et al., 2020)"

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(Magalhaes et al., 2021)

Enteric nervous system and inflammatory bowel diseases: Correlated impacts and therapeutic approaches through the P2X7 receptor

"As proposed by Aleksandr S. Dogiel in 1899, the morphological classification of enteric neurons can be based on their conformation and dendritic distribution. Dogiel described type I cells as flattened, slightly elongated, with an angled or star-shaped contour, and, as remarkable characteristics, as having only one axon and four to 20 Lamellar dendrites that frequently extend at a short distance from the cell body [31]"

"Type II neurons have large round or oval cell bodies and eccentric nuclei (Pompolo et al., 1988), and the surface is grooved by bundles of neural fibers (Hendriks et al., 1990). The main characteristic of type II neurons is the presence of several axonal processes that are emitted either directly from the cell body (multipolar neuron) or from a single initial process that branches into short subsidiary axons (pseudounipolar neurons) (Furness, 2012)(Song et al., 1994)"

"Additionally, enteric neurons can also be identified as intrinsic primary afferent neurons (IPANs), interneurons, and motor neurons

(Furness, 2012), classified into at least 18 subtypes and using more than 30 neurotransmitters in their synapses [28]30]. Of these neurotransmitters, acetylcholine (ACh) and nitric oxide (NO) stand out as the most abundant (Brookes, 2001), as well as adenosine-5'-triphosphate (ATP) (Hansen, 2003), vasoactive intestinal polypeptide (VIP), and substance P (SP) (Gallego et al., 2016)"

"IPANs (classified as Dogiel type II) are recognized for responding to chemical stimuli, mucosal deformation and GI muscle tension, translating these signals into a neural impulse that will trigger a local motor reflex (Costa et al., 2000). Altogether, IPANs represent approximately 14% and 30% of the neurons of the submucosal and myenteric plexuses, respectively"

"The enteric neural circuit is arranged in two plexuses: the submucosal plexus, which in large mammals is present in two individualized levels (outer/inner) and is located in the outer connective tissue layer and the inner mucosal layer, and the myenteric plexus, located between the longitudinal and circular muscle layers (Bayliss et al., 1899)(Furness, 2012)5,28]"

(Ren et al., 2008)

[Purinergic receptors and synaptic transmission in enteric neurons]

"There are enteric neurons that are sensitive to chemical or mechanical cues from the gastrointestinal tract with cell bodies in the myenteric and submucous plexes. They are properly referred to as intrinsic primary afferent neurons (IPANs) and are separate from the extrinsic primary afferent neurons whose cell bodies are in dorsal root or nodose ganglia (Furness et al., 1998). In this review we will simply refer to them as sensory neurons. Many enteric sensory neurons have "AH"-type electrophysiological characteristics which are defined as having an action potential afterhyperpolarization lasting >1 s (Hirst et al., 1974)(Bornstein et al., 1994)(Mao et al., 2006). Most sensory neurons have smooth cell bodies and are multipolar with projections that run to the mucosal epithelium, between the myenteric plexus and the submucosal plexus, and to their own and nearby ganglia (Pompolo et al., 1988). They form synapses with all other neuronal types"

"Enteric interneurons are primarily found in the myenteric plexus where there is generally one subtype of interneuron (two types in proximal colon) with an ascending projection and at least three subtypes of interneuron with a descending projection. There are also a small number of intestinofugal neurons that send a projection to prevertebral ganglia (Furness et al., 2002). All interneurons have a single axon, but there is a mix of dendritic morphology. Interneurons are characterized electrophysiologically as "S" neurons"

"In the myenteric plexus, ATP is utilized as a transmitter or co-transmitter by at least one type of descending interneuron during local reflexes (Bornstein, 2008)[17]. These descending interneurons are immunoreactive for nitric oxide synthase (NOS) and vasoactive intestinal polypeptide (VIP) and have Dogiel type I dendrites"

"Enteric motor neurons are found in the myenteric and submucous plexes. They have a single axon and short dendrites with Dogiel type I morphology. In the myenteric plexus, there are four functional classes of motor neuron to the smooth muscle layers of the intestine (Furness et al., 2002)(Brookes, 2001): inhibitory or excitatory motor neurons to either the circular or longitudinal smooth muscle layers. All motor neurons have S-type properties"

"In the myenteric plexus, inhibitory motor neurons to the circular muscle are immunoreactive for NOS and VIP and use ATP as a coneurotransmitter (Furness et al., 2002)(Costa et al., 1996)(Crist et al., 1992). In the submucous plexus, there are two types of excitatory secretomotor neurons going to the epithelium and at least one type of vasodilator neuron to the submucous arterioles. Noncholinergic secretomotor/ vasodilator neurons are immunoreactive for dynorphin, galanin, and VIP. These neurons account for ~45% of submucosal neurons, and they may utilize ATP as a coneurotransmitter (Monro et al., 2004)(Hu et al., 2003)."

(Zhao et al., 2023)

[The influence of Nav1.9 channels on intestinal hyperpathia and dysmotility]

"Neurons in the ENS are mainly divided into Dogiel type I and Dogiel type II neurons in morphology. The cell bodies of Dogiel type I neurons project multiple short dendrites and a single elongated axon. Dogiel type II neurons have smooth cell bodies and project multiple axons, but no dendrites or only a few dendrites [70]. In addition, ENS neurons are divided into sensory neurons, interneurons and motor neurons in terms of functional characteristics [71], and AH-type neurons and S-type neurons in terms of electrophysiological characteristics [72]. AH-type neurons are characterized by the larger action potentials with an infection on the falling phase and long afterhyperpolarizing potential (AHP;>2 s) that follows. S-type neurons are typified by the brief action potentials without slow AHPs, and they present fast excitatory postsynaptic potentials (EPSPs) [68]. The intrinsic sensory neurons/ intrinsic primary afferent neurons (IPANs) of the ENS generally have the characteristics of Dogiel II neurons in morphology, and belong to AH neurons in electrophysiological characteristics [67,73,74]. The ENS comprises the myenteric plexus and the submucosal plexus. The myenteric plexus is mainly responsible for the regulation of intestinal movements. The submucosal plexus is involved in the secretion of water and electrolytes, as well as the neural regulation of intestinal blood flow [68]."

(Bubeck et al., 2023)

[Guardians of the gut: influence of the enteric nervous system on the intestinal epithelial barrier]

"excitatory (PEMN/eMN) and inhibitory (PIMN/iMN) motor neurons innervate both the circular and longitudinal muscles and are connected by interneurons (PIN/IN). Sensory neurons (PSN) extend through the circular muscle layer, innervating the submucosal plexus as well as the epithelial layer. With the advent of single cell transcriptomic technologies four major studies have reported the

single cell transcriptomes of the human and mouse ENS. Following clustering and annotation of neuronal function, combined with cluster-labelling gene expression patterns, there have emerged seven distinct subtypes"

"The submucosal plexus, located beneath the circular muscle layer, primarily innervates the submucosal layer and the surrounding crypts of the enteric epithelium. It receives input from primary afferent neurons (IPAN) or secretomotor/vasodilator neurons (PSVN) and exhibits a more compact network with fewer cell bodies compared to the myenteric plexus. The main function of the submucosal plexus is to regulate secretory activity (Timmermans et al., 1997)."

(Makowska et al., 2019)

Age and Sex-Dependent Differences in the Neurochemical Characterization of Calcitonin Gene-Related Peptide-Like Immunoreactive (CGRP-LI) Nervous Structures in the Porcine Descending Colon

"The enteric plexuses consist of millions of neurons, which play various roles and express a range of neurotransmitters and/or neuromodulators (Furness et al., 2014)"

"Depending on the animal species and the fragment of the GI tract, the ENS structure shows some differences. In the esophagus and stomach, the enteric neurons of large mammals (e.g., pigs) are grouped into two ganglionated plexuses: the myenteric plexus (MP)-located between longitudinal and circular muscle layers and the submucous plexus placed near the lamina propria of the mucosal layer. In the small and large intestines, the submucous plexus is additionally divided into the outer submucous plexus (OSP) and inner submucous plexus (ISP), which are placed in the submucosal layer: the OSP near the inner side of the circular muscle layer and the ISP near the lamina propria of the mucosa (Balemba et al., 1998)(Petto et al., 2015)(Balemba et al., 1999)"

"enteric neurons are very diverse in terms of morphological, functional and electrophysiological properties, but the most important criterion of neuronal classification in the ENS is the neurochemical characterization of nerve cells (Balemba et al., 1998)(Petto et al., 2015). To date, several dozen neuronal active substances have been found in the enteric neurons. They may play various functions and are involved in all aspects of gastrointestinal physiology, such as motility, excretive activity, intestinal blood flow and ion transport [1]6]."

(Nikolla et al., 2025)

The Enteric Neuronal Circuitry: A Key Ignored Player in Nutrient Sensing Along the Gut–Brain Axis

"The ENS is comprised of the myenteric (Auerbach) plexus, nestled between the longitudinal and circular smooth muscle layers, and the submucosal (Meissner) plexus [1], embedded in the submucosal layer of the small intestine that collectively maintain gut homeostasis (Rubin et al., 2009). The myenteric plexus consists of myenteric ganglia of varying sizes. The neurons of the myenteric plexus innervate the muscularis externa and neighboring myenteric neurons (Gabella, 1979). Intrinsic primary afferent neurons (IPANs) have cell bodies in the myenteric plexus (Costa et al., 2000). Additionally, there are intestinofugal neurons (IFANs), a smaller population of myenteric neurons that project to the gallbladder, pancreas, and central nervous system (CNS) (Gabella, 1979). The submucosal plexus has two layers: an outer layer containing motor neurons that project to the smooth muscular layer and an inner layer that innervates the muscularis mucosae, defining the boundary between the mucosa and submucosa (Nezami et al., 2010)"

"Enteric neurons are typically identified and categorized based on morphology, location, chemistry, projections, and functions (Table 1). Historically, neurons have been classified as Dogiel type I or type II across a broad spectrum of species, with type III being the least understood type in humans [8]."

(Savulescu-Fiedler et al., 2025)

Rewiring the Brain Through the Gut: Insights into Microbiota–Nervous System Interactions

"The ENS is structured with two main plexuses represented by the submucous plexus (Meissner) and the myenteric plexus (Auerbach)"

"Both plexuses consist of three neuron types: the enteric intrinsic primary afferent neurons (IPANs), the motor neurons or efferent neurons, and the association neurons"

"IPANs represent the most vulnerable ENS neuronal population, given the fact that they extend to the lamina propria of the intestine (Benarroch, 2007)(Furness et al., 2004)"

"The enteric peristalsis is controlled by the motor neurons arising from the myenteric plexus, while absorption and secretion are regulated by the neurons emerging from the submucosal plexus (Furness et al., 2014)"

"The main classes of enteric neurons are represented by neurons that receive fast synaptic input, known as "S" neurons, and neurons with prolonged post-hyperpolarization, referred to as "AH" neurons (Natale et al., 2021). IPANs belong to "AH" neurons, whereas motor and interneurons are "S" neurons, which are relatively depolarized at rest [69]."

(Hao et al., 2009)

Development of enteric neuron diversity

> *"Initially, enteric neuron subtypes were identified by Dogiel based on morphological characteristics alone (see [1]). However, elegant studies performed in the past few decades utilizing a variety of immunohistochemical, electrophysiological, pharmacological and tracing techniques have identified different subtypes of enteric neurons based on a combination of morphology, neurotransmitters, electrophysiology, target tissue and direction (up, down or circumferentially) and length of axon projection.*
>
> *Enteric neurons have been most thoroughly studied in the guinea-pig ileum, where 10-15 subtypes of myenteric neurons and 4-5 subtypes of submucosal neurons have been identified; these include intrinsic primary afferent (sensory) neurons, ascending and descending interneurons, excitatory and inhibitory motor neurons to both the longitudinal and circular muscle layers, intestinofugal, secretomotor and vasodilator neurons [5][6][7]"*
>
> *"In mammals and birds, most enteric neurons are clustered in myenteric ganglia (Fig. 1A), which are located between the circular and longitudinal muscle layers, and in submucosal ganglia. In the intestine of smaller mammals, myenteric neurons are primarily involved in the regulation of gut motility and submucosal neurons are mostly involved in the regulation of secretion and vascular tone; in larger mammals, some submucosal neurons are also directly involved in motility reflexes [4]."*

(Zeisel et al., 2018)

Molecular Architecture of the Mouse Nervous System

> *"The mammalian nervous system executes complex behaviors controlled by specialised, precisely positioned and interacting cell types. Here, we used RNA sequencing of half a million single cells to create a detailed census of cell types in the mouse nervous system. We mapped cell types spatially and derived a hierarchical, data-driven taxonomy. Neurons were the most diverse, and were grouped by developmental anatomical units, and by the expression of neurotransmitters and neuropeptides. Neuronal diversity was driven by genes encoding cell identity, synaptic connectivity, neurotransmission and membrane conductance. We discovered several distinct, regionally restricted, astrocytes types, which obeyed developmental boundaries and correlated with the spatial distribution of key glutamate and glycine neurotransmitters. In contrast, oligodendrocytes showed a loss of regional identity, followed by a secondary diversification. The resource presented here lays a solid foundation for understanding the molecular architecture of the mammalian nervous system, and enables genetic manipulation of specific cell types."*

(Llorente, 2024)

The Imperative for Innovative Enteric Nervous System–Intestinal Organoid Co-Culture Models: Transforming GI Disease Modeling and Treatment

> *"Immunohistochemistry, morphological, and single-cell RNA sequencing (scRNAseq) analyses have been instrumental in the precise classification of enteric neuron subtypes. Through these techniques, studies have elucidated the primary neurotransmitters that delineate various neuron types within the myenteric plexus. While most myenteric neurons are traditionally classified as either cholinergic or nitrergic, research has unveiled complexities, with some neurons exhibiting dual characteristics, lacking both, or expressing glial markers. These facts challenge the established paradigms and underscore the criticality of accurate classification when exploring enteric neuronal populations [116][117][118]. Submucosal neurons, on the other hand, mainly consist of cholinergic neurons and vasoactive intestinal peptide (VIP)-expressing noncholinergic neurons. These major neuronal populations can further divide into subsets based on additional markers like neuropeptides and calcium-binding proteins [4]. The proportions of these subsets vary along the gut and show interspecies differences. Overall, around 15 classes of functionally defined, neurochemically coded enteric neurons have been identified in the intestine, with fewer in the stomach and esophagus [2]."*

Transcriptomic Classifications and Single-Cell Studies

Single-cell RNA sequencing has revolutionized enteric neuron classification by revealing 12 distinct molecular classes in the mouse myenteric plexus, providing unprecedented resolution of neuronal diversity and enabling the identification of novel marker genes and regulatory pathways that define specific neuronal subtypes. These transcriptomic studies have validated traditional functional classifications while uncovering previously unknown molecular complexity and species-specific differences in gene expression patterns. (8 sources)

The advent of single-cell RNA sequencing (scRNA-seq) has transformed our understanding of enteric neuron diversity by enabling comprehensive transcriptional profiling of individual neurons within the ENS (Morarach et al., 2020) (Benthal et al., 2026). These technological advances have moved beyond traditional classification methods that relied on morphology, immunohistochemistry, and electrophysiology to provide unprecedented molecular resolution of neuronal subtypes (Majd et al., 2024).

Molecular taxonomy of enteric neurons has been most comprehensively defined through large-scale single-cell studies that have profiled thousands of individual neurons. The most detailed classification identified 12 distinct enteric neuron classes (ENCs) within the myenteric plexus of the mouse small intestine, with some classes defined by selective expression of single genes including somatostatin (Sst), neuromedin U (Nmu), cholecystokinin (Cck), urocortin 3 (Ucn3), neuronal differentiation 6 (Neurod6), and neurexophilin-2 (Nxph2) (Morarach et al., 2020). Other classes are distinguished by combinatorial gene codes, such as ENC1-4 displaying shared expression of Tac1 but differing in Calb2, Ndufa4l2, Gda, Penk and Fut9 expression (Morarach et al., 2020).

Broad transcriptomic categories align closely with classical neurotransmitter-based classifications. Single-cell sequencing reveals that enteric neurons can be grouped into two main populations based on primary neurotransmitters: nitric oxide synthase 1 (NOS1) expressing neurons and choline acetyltransferase (ChAT) expressing neurons (Jakob et al., 2020) (Zeisel et al., 2018). NOS1+ neurons comprise three distinct clusters representing inhibitory motor neurons and secretomotor/vasodilator neurons, while ChAT+ neurons include six clusters encompassing excitatory motor neurons, sensory neurons, and interneurons (Jakob et al., 2021) (Zeisel et al., 2018).

Functional correlation with transcriptomic data has validated traditional classifications while providing molecular precision. ENC1-4 were assigned to excitatory motor neurons based on Tac1/Calb2 expression, while ENC8 and 9 matched inhibitory motor neurons expressing Nos1/Gal/Vip/Npy (Morarach et al., 2020). IPANs were identified as ENC6, 7, and 12, expressing different combinations of previously defined markers (Calca/Calcb/Nfem/Calb1/Calb2) and clearly separable by selective expression of Nmu, Ucn-3/Cck, or Nxph2 (Morarach et al., 2020).

Comparative studies between humans and mice have revealed both conserved and species-specific transcriptional programs. Large-scale profiling of over 25,000 mouse neurons and 1,445 human neurons showed that several enteric neuron subtypes in the duodenum, ileum, and colon are conserved between species based on orthologous gene expression (Bianco et al., 2021) (May-Zhang et al., 2020). However, some enteric neuron subtype-specific genes in mice are expressed in completely distinct morphologically defined neuron subtypes in humans, highlighting important species-specific differences that must be considered when translating findings from mouse to human ENS (Bianco et al., 2021) (May-Zhang et al., 2020).

Technical advances and integration challenges continue to refine transcriptomic classifications. Multiple independent datasets are being computationally integrated to identify new cell types, shared marker genes across developmental stages, and achieve consensus on transcriptomic definitions of enteric neuronal subtypes (Benthal et al., 2026). However, a significant challenge remains in connecting these deep transcriptional profiles with historical classification systems based on morphology and function (Benthal et al., 2026). Despite decades of functional characterization identifying major cell types using markers like NOS1 for inhibitory motor neurons, acetylcholine for excitatory motor neurons, and CALCA/B for IPANs, single-cell studies are revealing yet-to-be-determined factors that confer cell-type specific functional properties including ion channels, metabolic profiles, and specialized capabilities for cell-cell interaction (Majd et al., 2024).

Evidence

(Morarach et al., 2020)

Diversification of molecularly defined myenteric neuron classes revealed by single cell RNA-sequencing

> *"Meticulous work over the last decades has identified a framework of enteric neurons with sensory (Intrinsic Primary Afferent Neurons, IPANs), motor or interneuron characteristics"*
>
> *"we define a molecular taxonomy of 12 enteric neuron classes within the myenteric plexus of the mouse small intestine using single-cell RNA sequencing. We present cell–cell communication features and histochemical markers for motor neurons, sensory neurons and interneurons"*
>
> *"some classes were defined by the selective expression of single genes with likely importance for their neuronal activity. These included ENC5 (somatostatin; Sst), ENC6 (neuromedin U; Nmu), ENC7 (cholecystokinin; Cck and urocortin 3; Ucn3), ENC10 (neuronal differentiation 6; Neurod6) and ENC12 (neurexophilin-2; Nxph2). Other classes were distinguished by combinatorial gene codes"*

"ENC1-4 displayed a partly shared gene expression, including Tac1, but differed in their expression of Calb2, Ndufa4l2, Gda, Penk and Fut9"

"ENC8-9 jointly expressed Nos1 and c1ql1, but Npy and Cox8b were significantly lower in ENC9. ENC11 highly expressed Npy, but lacked Nos1"

"ENC1-4 were assigned to excitatory motor neurons (Tac1/Calb2), while ENC8 and 9 matched inhibitory motor neurons (Nos1/Gal/Vip/Npy). ENC6, 7 and 12 were mapped to IPANs and expressed different combinations of previously defined markers (Calca/Calcb/Nfem/Calb1/Calb2). These plausible IPANs were clearly separable by their selective expression of Nmu, Ucn-3/Cck or Nxph2."

(Benthal et al., 2026)

Building consensus: construction of a juvenile and adult scRNA-seq meta-atlas for dataset comparisons and harmonizing transcriptomic definitions of enteric neuron subtypes

"Previously, distinct enteric neuron types were cataloged by dye-filling techniques, immunohistochemistry, retrograde labeling, and electrophysiology. Recent technical advances in single cell RNA-sequencing (scRNA-seq) have enabled transcriptional profiling of hundreds to millions of individual cells from the intestine. These data allow cell types to be resolved and compared using their transcriptional profiles ("clusters") rather than relying on antibody labeling"

"Here we briefly systematically review each Mus musculus single cell or single nucleus RNA-sequencing (snRNA-seq) ENS dataset. We then reprocess and computationally integrate these select independent scRNA-seq enteric neuron datasets from the myenteric plexus with the aim to identify new cell types, shared marker genes across juvenile to adult ages, dataset differences, and achieve some consensus on transcriptomic definitions of enteric neuronal subtypes"

"Future studies face the challenge of connecting these deep transcriptional profiles for enteric neurons with historical classification systems."

(Majd et al., 2024)

A call for a unified and multimodal definition of cellular identity in the enteric nervous system

"For decades, enteric neuron subtype identity has largely been defined based on morphological, immunohistochemical and functional features of neurons within the gut, including neuronal projection localization, neurochemistry, and electrophysiological properties. Investigations conducted on tissues isolated from model organisms, such as the guinea pig, mouse and rat 1 have significantly progressed our understanding of the neurochemical and functional complexity of enteric neuron subtypes and have identified functional units within the ENS circuitry. These include intrinsic primary afferent neurons (IPANs or sensory neurons, SNs), ascending and descending interneurons (IN), excitatory and inhibitory motor neurons (EMNs and IMNs), and secretomotor and/or vasodilator neurons (SVNs) 1 . Despite the fact that many known neurotransmitters and neuropeptides are present in the ENS, only a handful are consistently used as markers to identify major cell types in the ENS. Some examples include nitric oxide synthase (NOS1) for IMNs and acetylcholine for EMNs as well as CALCA/B for IPANs 2 . While it is impressive how large swaths of ENS function are coordinated by these neurotransmitters and neuropeptides, it also raises the question of whether yet-to-be-determined factors confer cell-type specific functional properties 3 . These features might include secreted peptides and proteins, electrophysiological features mediated by ion channels and transporters, metabolic profiles, and specialized capabilities for cell-cell interaction or synapse formation. To accurately understand the identity of individual cell types, it is also crucial to consider the contextual information in which they operate, including their position in the gut. Histochemical and functional assays have identified differences along the length of the GI tract, as well as between species, highlighting the need for more systematic efforts towards comprehensive mapping and profiling of the ENS 1 ."

(Jakob et al., 2020)

Neuro-Immune Circuits Regulate Immune Responses in Tissues and Organ Homeostasis

"The ENS is organized in plexuses throughout the intestine composed of neurons whose cell bodies lie within the intestinal wall (Figure 2)"

"The myenteric plexus (Auerbach plexus) lies between the longitudinal and circular muscle layer in the intestinal wall whereas the submucosal plexus (Meissner plexus) forms a network within the submucosal layer (Figure 2)"

"Neurons in the ENS can be categorized based on their anatomy, function and neurotransmitter signature. Up to 20 functional classes of neurons can be identified in the guinea pig. Functionally and phenotypically, several types of enteric neurons are distinguished and can be further sub-classified: Excitatory neurons innervating intestinal muscles, inhibitory neurons innervating intestinal muscles (to circular and longitudinal muscles, respectively), secretomotor and vasodilator neurons, secretomotor neurons without vasodilator activity and neurons to enteroendocrine cells, sensory intrinsic primary afferent neurons, ascending and descending interneurons and intestinofugal neurons (Furness, 2000)(Furness, 2006)(Furness, 2008). Single-cell sequencing experiments revealed nine clusters of enteric neurons in mice, which can be classified based on the two neurotransmitters nitric oxide (NO) (Nos1 expression, cluster 1-3) and acetylcholine (Chat expression, cluster 4-9) (Zeisel et al., 2018). Structurally, Nos1 + neurons are preferentially type I neurons whereas among Chat + neurons type II neurons are overrepresented (34)."

(Zeisel et al., 2018)

Molecular Architecture of the Mouse Nervous System

"The mammalian nervous system executes complex behaviors controlled by specialised, precisely positioned and interacting cell types. Here, we used RNA sequencing of half a million single cells to create a detailed census of cell types in the mouse nervous system. We mapped cell types spatially and derived a hierarchical, data-driven taxonomy. Neurons were the most diverse, and were grouped by developmental anatomical units, and by the expression of neurotransmitters and neuropeptides. Neuronal diversity was driven by genes encoding cell identity, synaptic connectivity, neurotransmission and membrane conductance. We discovered several distinct, regionally restricted, astrocytes types, which obeyed developmental boundaries and correlated with the spatial distribution of key glutamate and glycine neurotransmitters. In contrast, oligodendrocytes showed a loss of regional identity, followed by a secondary diversification. The resource presented here lays a solid foundation for understanding the molecular architecture of the mammalian nervous system, and enables genetic manipulation of specific cell types."

(Jakob et al., 2021)

An Integrated View on Neuronal Subsets in the Peripheral Nervous System and Their Role in Immunoregulation

"The somas of enteric neurons mainly cluster in two anatomically distinct but strongly interconnected ganglionated plexuses (10). The outer, myenteric plexus (Auerbach Plexus) lies within the longitudinal and the circular muscle layer, and the inner, submucosal plexus (Meissner Plexus) is located below the muscle layers (Jakob et al., 2020)"

"The intrinsic micro-circuitry regulation of intestinal function consists of five neuronal classes with distinct functional specializations. Sensory neurons, also referred to as intrinsic primary afferent neurons (IPANs), detect chemical and physical alterations in the intestine and transmit the signal via interneurons to excitatory motor neurons, inhibitory motor neurons or secretomotor/vasodilator neurons"

"Single-cell sequencing of the ENS using fluorescenceactivated sorting of Wnt1-Cre; R26Tomato mice revealed 1105 enteric neurons (with ~3066 genes detected on average) and eventually found nine clusters of neurons in the muscular sheet of the intestine (Zeisel et al., 2018). Based on neuro-transmitters nitric oxide synthase 1 (NOS1) and choline acetyltransferase (ChAT), two main groups can be classified, NOS1 + neurons and ChAT + neurons. The authors assigned three distinct neuronal clusters of nitrergic neurons, which comprise inhibitory motor neurons and secretomotor/vasodilator neurons (Table 1). Further, 6 clusters of cholinergic neurons, which express ChAT and the solute carrier family 5 member 7 (Slc5a7, gene encodes for an ion transporter) could be distinguished (Zeisel et al., 2018). ChAT + neurons are further subdivided into excitatory motor neurons, sensory neurons and interneurons according to their gene expression (Table 1)"

"Using the pan-neuronal Baf53b-cre deleter mice crossed to R26R-tomato to enable sort-purification of enteric neurons from the myenteric plexus of the small intestine combined with singlecell RNA-sequencing, 4,892 high-quality enteric neurons have been analyzed and 12 classes of neurons could be identified based on their gene expression (13)."

(Bianco et al., 2021)

Novel understanding on genetic mechanisms of enteric neuropathies leading to severe gut dysmotility

"Emerging research has provided novel insights on marker genes specific for the different enteric neuron classes (ENCs). Recently, Morarach et al. (Morarach et al., 2020) addressed this extremely innovative area and provided molecular evidence of twelve ENCs within the myenteric plexus of the mouse small intestine. The authors identified inter-communication features and histochemical markers for discrete classes of neurons, including motor, sensory, and interneurons together with genetic tools for class-specific targeting. (Morarach et al., 2020) A recent work from Ganz et al. (Ganz et al., 2019) showed that genes controlling epigenetic modifications are important to coordinate intestinal tract development, providing the first demonstration that these genes influence ENS development. In this line, another study in human and mouse healthy tissues used laser-capture microdissection coupled to single-nucleus RNA-sequencing, to map out enteric neuron subtypes in the duodenum, ileum, and colon, which were overall conserved between humans and mice based on orthologous gene expression. Nevertheless, some enteric neuron subtype-specific genes in mice were expressed in completely distinct morphologically defined neuron subtypes in humans, thus suggesting that these species-specific differences should be taken into account when translating findings from mouse to human ENS (May-Zhang et al., 2020) ."

(May-Zhang et al., 2020)

Combinatorial transcriptional profiling of mouse and human enteric neurons identifies shared and disparate subtypes in situ

"BACKGROUND & AIMS The enteric nervous system (ENS) coordinates essential intestinal functions through the concerted action of diverse enteric neurons (EN). However, integrated molecular knowledge of EN subtypes is lacking. To compare human and mouse ENs, we transcriptionally profiled healthy ENS from adult humans and mice. We aimed to identify transcripts marking discrete neuron subtypes and visualize conserved EN subtypes for humans and mice in multiple bowel regions. METHODS Human myenteric ganglia and adjacent smooth muscle were isolated by laser-capture microdissection for RNA-Seq. Ganglia-specific transcriptional profiles were identified by computationally subtracting muscle gene signatures. Nuclei from mouse myenteric neurons were isolated and subjected to single-nucleus RNA-Seq (snRNA-Seq), totaling over four billion reads and 25,208 neurons. Neuronal subtypes were

defined using mouse snRNA-Seq data. Comparative informatics between human and mouse datasets identified shared EN subtype markers, which were visualized in situ using hybridization chain reaction (HCR). RESULTS Several EN subtypes in the duodenum, ileum, and colon are conserved between humans and mice based on orthologous gene expression. However, some EN subtype-specific genes from mice are expressed in completely distinct morphologically defined subtypes in humans. In mice, we identified several neuronal subtypes that stably express gene modules across all intestinal segments, with graded, regional expression of one or more marker genes. CONCLUSIONS Our combined transcriptional profiling of human myenteric ganglia and mouse EN provides a rich foundation for developing novel intestinal therapeutics. There is congruency among some EN subtypes, but we note multiple species differences that should be carefully considered when relating findings from mouse ENS research to human GI studies. Graphical Abstract"

Species-Specific Differences and Comparative Studies

While fundamental ENS organization and major neuronal classes are conserved across mammalian species, significant differences exist in neurochemical coding, regional distribution, and gene expression patterns between species, particularly between rodent models and humans. These species-specific variations must be carefully considered when translating findings from laboratory animals to human gastrointestinal disorders. (10 sources)

The remarkable conservation of basic ENS architecture across mammalian species has enabled decades of productive research using animal models, yet important species-specific differences in neuronal organization, neurochemical coding, and molecular expression patterns have emerged from comparative studies (Dharshika et al., 2022). The core features of enteric neuron subtypes, including the fundamental distinction between excitatory motor neurons, inhibitory motor neurons, interneurons, and sensory neurons, are conserved between mice and humans, providing a solid foundation for translational research (Dharshika et al., 2022).

Molecular conservation and divergence between species has been most comprehensively examined through large-scale transcriptomic studies. Several enteric neuron subtypes in the duodenum, ileum, and colon show conservation between humans and mice based on orthologous gene expression patterns (Bianco et al., 2021) (May-Zhang et al., 2020). However, critical differences exist where some enteric neuron subtype-specific genes in mice are expressed in completely distinct morphologically defined neuron subtypes in humans, highlighting the importance of species-specific validation when translating findings from mouse to human ENS (Bianco et al., 2021) (May-Zhang et al., 2020).

Regional and neurochemical variations show significant species-dependent patterns. The anatomical organization of enteric plexuses varies across species, with larger mammals like pigs showing more complex submucosal organization compared to smaller animals (Makowska et al., 2019). In the porcine colon, the submucosal plexus displays distinct anatomical and neurochemical features, with the inner submucosal plexus containing predominantly cholinergic neurons and the outer submucosal plexus showing more complex neurochemical coding including both cholinergic and nitrergic populations (Makowska et al., 2019) (Petto et al., 2015).

Dopaminergic neuron distributions illustrate striking species differences in neuronal proportions and regional organization. In mice, dopaminergic neurons account for approximately 10-13% of enteric neurons, while in humans this proportion can reach up to 20%, with both species showing an oral-to-aboral gradient where these neurons are more prevalent in the upper gastrointestinal tract (Pradeloux et al., 2025) (Li et al., 2003). Expression of dopaminergic markers like tyrosine hydroxylase shows regional variability, being more prominent in the stomach and ileum compared to the colon in mice (Pradeloux et al., 2025).

Developmental and functional considerations reveal that approximately 15 classes of functionally defined, neurochemically coded enteric neurons have been identified in the intestine across species, with fewer classes found in the stomach and esophagus (Llorente, 2024) (Parathan et al., 2020). The proportions of neuronal subsets vary along the gut and show interspecies differences, with some neurons exhibiting dual neurotransmitter characteristics or lacking traditional markers entirely, challenging established classification paradigms (Llorente, 2024).

Implications for translational research emphasize the need for systematic comparative mapping efforts across species. While decades of functional characterization in model organisms have identified major cell types using conserved markers like nitric oxide synthase for inhibitory motor neurons and acetylcholine for excitatory motor neurons, the discovery of species-specific differences in gene expression patterns requires careful validation when applying findings to human gastrointestinal disorders (Majd et al., 2024). These comparative studies highlight the importance of integrating histochemical, functional, and molecular approaches to accurately map ENS diversity across species and gut regions (Majd et al., 2024).

Evidence

(Dharshika et al., 2022)

Enteric Neuromics: How High-Throughput “Omics” Deepens Our Understanding of Enteric Nervous System Genetic Architecture

> *"Type I neurons have flat, elongate, and irregular cell bodies with a single axon and numerous short dendrites, whereas type II neurons have smoother and larger cell bodies with multiple long axons. Type I neurons project and communicate with adjacent ganglia in the plexus and musculature, whereas type II neurons communicate with neurons throughout the gut wall, within and between ganglia, and the mucosa"*

> *"are also classified by electrophysiological properties, which have been mainly characterized in guinea pigs and mice. Two main types of enteric neurons are categorized as having either synaptic-or AH (after hyperpolarization)-type electrophysiological properties that differ on the basis of action potential speed and magnitude, the length of AH potentials, and tetrodotoxin sensitivity. Synaptic-type neurons typically display Dogiel type I morphology and include interneurons and motor neurons, whereas AH-type neurons typically display Dogiel type II morphology and are considered sensory neurons"*

> *"Defining the neurochemical coding of enteric neurons was a significant advancement in identifying neuronal subtypes and understanding how enteric neurons communicate with one another and target cells. Enteric neurochemical coding has been defined by multiple approaches including immunohistochemistry in combination with retrograde tracing, electrophysiology, and pharmacology. Integrating these biomolecular data with morphologic and electrophysiological properties is the basis for current definitions of enteric neuron subtypes, which include motor neurons, interneurons, and sensory neurons. Although these definitions are based largely on studies in guinea pigs, many of the core features of enteric neuron subtypes are conserved between mice and humans. Excitatory and inhibitory motor neurons reside in the myenteric plexus and innervate the circular and longitudinal muscle of the intestine."*

(Bianco et al., 2021)

Novel understanding on genetic mechanisms of enteric neuropathies leading to severe gut dysmotility

> *"Emerging research has provided novel insights on marker genes specific for the different enteric neuron classes (ENCs). Recently, Morarach et al. (Morarach et al., 2020) addressed this extremely innovative area and provided molecular evidence of twelve ENCs within the myenteric plexus of the mouse small intestine. The authors identified inter-communication features and histochemical markers for discrete classes of neurons, including motor, sensory, and interneurons together with genetic tools for class-specific targeting. (Morarach et al., 2020) A recent work from Ganz et al. (Ganz et al., 2019) showed that genes controlling epigenetic modifications are important to coordinate intestinal tract development, providing the first demonstration that these genes influence ENS development. In this line, another study in human and mouse healthy tissues used laser-capture microdissection coupled to single-nucleus RNA-sequencing, to map out enteric neuron subtypes in the duodenum, ileum, and colon, which were overall conserved between humans and mice based on orthologous gene expression. Nevertheless, some enteric neuron subtype-specific genes in mice were expressed in completely distinct morphologically defined neuron subtypes in humans, thus suggesting that these species-specific differences should be taken into account when translating findings from mouse to human ENS (May-Zhang et al., 2020) ."*

(May-Zhang et al., 2020)

Combinatorial transcriptional profiling of mouse and human enteric neurons identifies shared and disparate subtypes in situ

> *"BACKGROUND & AIMS The enteric nervous system (ENS) coordinates essential intestinal functions through the concerted action of diverse enteric neurons (EN). However, integrated molecular knowledge of EN subtypes is lacking. To compare human and mouse ENs, we transcriptionally profiled healthy ENS from adult humans and mice. We aimed to identify transcripts marking discrete neuron subtypes and visualize conserved EN subtypes for humans and mice in multiple bowel regions. METHODS Human myenteric ganglia and adjacent smooth muscle were isolated by laser-capture microdissection for RNA-Seq. Ganglia-specific transcriptional profiles were identified by computationally subtracting muscle gene signatures. Nuclei from mouse myenteric neurons were isolated and subjected to single-nucleus RNA-Seq (snRNA-Seq), totaling over four billion reads and 25,208 neurons. Neuronal subtypes were defined using mouse snRNA-Seq data. Comparative informatics between human and mouse datasets identified shared EN subtype markers, which were visualized in situ using hybridization chain reaction (HCR). RESULTS Several EN subtypes in the duodenum, ileum, and colon are conserved between humans and mice based on orthologous gene expression. However, some EN subtype-specific genes from mice are expressed in completely distinct morphologically defined subtypes in humans. In mice, we identified*

several neuronal subtypes that stably express gene modules across all intestinal segments, with graded, regional expression of one or more marker genes. CONCLUSIONS Our combined transcriptional profiling of human myenteric ganglia and mouse EN provides a rich foundation for developing novel intestinal therapeutics. There is congruency among some EN subtypes, but we note multiple species differences that should be carefully considered when relating findings from mouse ENS research to human GI studies. Graphical Abstract"

(Makowska et al., 2019)

[Age and Sex-Dependent Differences in the Neurochemical Characterization of Calcitonin Gene-Related Peptide-Like Immunoreactive (CGRP-LI) Nervous Structures in the Porcine Descending Colon]()

> *"The enteric plexuses consist of millions of neurons, which play various roles and express a range of neurotransmitters and/or neuromodulators (Furness et al., 2014)"*
>
> *"Depending on the animal species and the fragment of the GI tract, the ENS structure shows some differences. In the esophagus and stomach, the enteric neurons of large mammals (e.g., pigs) are grouped into two ganglionated plexuses: the myenteric plexus (MP)-located between longitudinal and circular muscle layers and the submucous plexus placed near the lamina propria of the mucosal layer. In the small and large intestines, the submucous plexus is additionally divided into the outer submucous plexus (OSP) and inner submucous plexus (ISP), which are placed in the submucosal layer: the OSP near the inner side of the circular muscle layer and the ISP near the lamina propria of the mucosa (Balemba et al., 1998)(Petto et al., 2015)(Balemba et al., 1999)"*
>
> *"enteric neurons are very diverse in terms of morphological, functional and electrophysiological properties, but the most important criterion of neuronal classification in the ENS is the neurochemical characterization of nerve cells (Balemba et al., 1998)(Petto et al., 2015). To date, several dozen neuronal active substances have been found in the enteric neurons. They may play various functions and are involved in all aspects of gastrointestinal physiology, such as motility, excretive activity, intestinal blood flow and ion transport [1]6]."*

(Petto et al., 2015)

[Architecture and Chemical Coding of the Inner and Outer Submucous Plexus in the Colon of Piglets]()

> *"In the porcine colon, the submucous plexus is divided into an inner submucous plexus (ISP) on the epithelial side and an outer submucous plexus (OSP) on the circular muscle side. Although both plexuses are probably involved in the regulation of epithelial functions, they might differ in function and neurochemical coding according to their localization. Therefore, we examined expression and co-localization of different neurotransmitters and neuronal markers in both plexuses as well as in neuronal fibres. Immunohistochemical staining was performed on wholemount preparations of ISP and OSP and on cryostat sections. Antibodies against choline acetyltransferase (ChAT), substance P (SP), somatostatin (SOM), neuropeptide Y (NPY), vasoactive intestinal peptide (VIP), neuronal nitric oxide synthase (nNOS) and the pan-neuronal markers Hu C/D and neuron specific enolase (NSE) were used. The ISP contained 1,380 ± 131 ganglia per cm2 and 122 ± 12 neurons per ganglion. In contrast, the OSP showed a wider meshwork (215 ± 33 ganglia per cm2) and smaller ganglia (57 ± 3 neurons per ganglion). In the ISP, 42% of all neurons expressed ChAT. About 66% of ChAT-positive neurons co-localized SP. A small number of ISP neurons expressed SOM. Chemical coding in the OSP was more complex. Besides the ChAT/±SP subpopulation (32% of all neurons), a nNOS-immunoreactive population (31%) was detected. Most nitrergic neurons were only immunoreactive for nNOS; 10% co-localized with VIP. A small subpopulation of OSP neurons was immunoreactive for ChAT/nNOS/±VIP. All types of neurotransmitters found in the ISP or OSP were also detected in neuronal fibres within the mucosa. We suppose that the cholinergic population in the ISP is involved in the control of epithelial functions. Regarding neurochemical coding, the OSP shares some similarities with the myenteric plexus. Because of its location and neurochemical characteristics, the OSP may be involved in controlling both the mucosa and circular muscle."*

(Pradeloux et al., 2025)

[Impact of aging on the central and enteric nervous system in a Parkinson’s disease mouse model]()

> *"Functionally, enteric neurons are typically classified into five major categories based on their neurotransmitter profiles:*
>
> *(1) excitatory motor neurons (cholinergic), (2) inhibitory motor neurons (nitrergic), (3) sensory neurons, (4) excitatory and inhibitory interneurons, and (5) secretomotor/vasodilator neurons (Natale et al., 2021). The cholinergic neurons make up the majority, at least 70% of myenteric neurons (Anlauf et al., 2003)"*
>
> *"Those neurons, constituting less than 1% of all enteric cells, are organized into ganglionic plexuses-primarily the myenteric (MyP) and submucosal (SMP) plexuses-or into non-ganglionic nerve fiber bundles (Furness et al., 1980)(Brudek, 2019)"*
>
> *"DAergic neurons represent a relatively small but functionally significant subset of the ENS. In mice, they account for approximately 10-13% of enteric neurons, while in humans, this proportion can reach up to 20%, with a distribution that follows an oral-to-aboral gradient (Anlauf et al., 2003)(Li et al., 2003). These neurons are particularly enriched in the upper gastrointestinal tract, where they constitute 14-20% of the neuronal population, but their prevalence drops to 1-6% in the distal small intestine. DAergic enteric neurons express key dopaminergic markers, including tyrosine hydroxylase (TH), dopamine transporter (DAT), and all five dopamine receptor subtypes (D1-D5)"*

"Expression of the dopaminergic phenotype in the mouse ENS is variable depending on the region of the gut, TH transcripts being more prominent in the stomach and ileum compared to the colon (Chalazonitis et al., 2022)."

(Li et al., 2003)

Enteric Dopaminergic Neurons: Definition, Developmental Lineage, and Effects of Extrinsic Denervation

"The existence of enteric dopaminergic neurons has been suspected; however, the innervation of the gut by sympathetic nerves, in which dopamine (DA) is the norepinephrine precursor, complicates analyses of enteric DA. We now report that transcripts encoding tyrosine hydroxylase (TH) and the DA transporter (DAT) are present in the murine bowel (small intestine > stomach or colon; proximal colon > distal colon). Because sympathetic neurons are extrinsic, transcripts encoding TH and DAT in the bowel are probably derived from intrinsic neurons. TH protein was demonstrated immunocytochemically in neuronal perikarya (submucosal >> myenteric plexus; small intestine > stomach or colon). TH, DA, and DAT immunoreactivities were coincident in subsets of neurons (submucosal > myenteric) in guinea pig and mouse intestines in situ and in cultured guinea pig enteric ganglia. Surgical ablation of sympathetic nerves by extrinsic denervation of loops of the bowel did not affect DAT immunoreactivity but actually increased numbers of TH-immunoreactive neurons, expression of mRNA encoding TH and DAT, and enteric DOPAC (the specific dopamine metabolite). The fetal gut contains transiently catecholaminergic (TC) cells. TC cells are the proliferating crest-derived precursors of mature neurons that are not catecholaminergic and, thus, disappear after embryonic day (E) 14 (mouse) or E15 (rat). TC cells appear early in ontogeny, and their development/survival is dependent on mash-1 gene expression. In contrast, the intrinsic TH-expressing neurons of the murine bowel appear late (perinatally) and are mash-1 independent. We conclude that the enteric nervous system contains intrinsic dopaminergic neurons that arise from a mash-1-independent lineage of noncatecholaminergic precursors."

(Llorente, 2024)

The Imperative for Innovative Enteric Nervous System–Intestinal Organoid Co-Culture Models: Transforming GI Disease Modeling and Treatment

"Immunohistochemistry, morphological, and single-cell RNA sequencing (scRNAseq) analyses have been instrumental in the precise classification of enteric neuron subtypes. Through these techniques, studies have elucidated the primary neurotransmitters that delineate various neuron types within the myenteric plexus. While most myenteric neurons are traditionally classified as either cholinergic or nitrergic, research has unveiled complexities, with some neurons exhibiting dual characteristics, lacking both, or expressing glial markers. These facts challenge the established paradigms and underscore the criticality of accurate classification when exploring enteric neuronal populations [116][117][118]. Submucosal neurons, on the other hand, mainly consist of cholinergic neurons and vasoactive intestinal peptide (VIP)-expressing noncholinergic neurons. These major neuronal populations can further divide into subsets based on additional markers like neuropeptides and calcium-binding proteins [4]. The proportions of these subsets vary along the gut and show interspecies differences. Overall, around 15 classes of functionally defined, neurochemically coded enteric neurons have been identified in the intestine, with fewer in the stomach and esophagus [2]."

(Parathan et al., 2020)

The enteric nervous system undergoes significant chemical and synaptic maturation during adolescence in mice.

"Adolescence is a critical period of development. It is very likely that there is significant maturation of the enteric nervous system (ENS) of the gut during this stage of life, especially since there are substantial changes in factors known to influence the ENS including diet and microbiota during this time, but this remains unknown. To examine maturation of the ENS during adolescence, we performed immunohistochemistry using advanced microscopy and analytical methods to compare enteric neurons and glia of the duodenum and colon of mice taken prior to weaning with those of young adult mice. We found significant changes in the architecture of both myenteric and submucosal plexuses and surprisingly found subsets of enteric cells that co-expressed the pan-neuronal marker, Hu, and either glial markers Sox10 or S100β, not both. About 70% and 35% of all Hu + neurons in the submucous plexus of the young adult duodenum and colon respectively also expressed S100β. The proportion of Hu+/Sox10 + cells in the duodenal myenteric plexus decreased, while the proportion of Hu+/S100β+ cells in the colonic submucosal plexus increased during adolescence. In the submucous plexus, there were significant increases in the proportions of vasoactive intestinal peptide+ and choline acetyltransferase + secretomotor neurons, of neurofilament M (NFM)+ neurons in the colon and of calretinin + neurons in the duodenum during adolescence. There were no age-dependent changes in the neurochemistry of various myenteric neuronal subtypes, including those immunoreactive for neuronal nitric oxide synthase (nNOS), Calbindin, Calretinin or NFM. There were significant increases in the somata sizes of Calretinin + submucosal and myenteric neurons, and nNOS + myenteric neurons, and these enteric neurons received significantly more synaptophysin + contacts onto their cell bodies during adolescence. This is the first study showing that enteric neurons and glia in the gut undergo significant changes in their anatomy and chemistry during adolescence. Notably changes in synaptic contacts within the enteric circuitry strongly suggest maturation in gastrointestinal function occurs during this time."

(Majd et al., 2024)

A call for a unified and multimodal definition of cellular identity in the enteric nervous system

"For decades, enteric neuron subtype identity has largely been defined based on morphological, immunohistochemical and functional features of neurons within the gut, including neuronal projection localization, neurochemistry, and electrophysiological properties. Investigations conducted on tissues isolated from model organisms, such as the guinea pig, mouse and rat 1 have

significantly progressed our understanding of the neurochemical and functional complexity of enteric neuron subtypes and have identified functional units within the ENS circuitry. These include intrinsic primary afferent neurons (IPANs or sensory neurons, SNs), ascending and descending interneurons (IN), excitatory and inhibitory motor neurons (EMNs and IMNs), and secretomotor and/or vasodilator neurons (SVNs) 1 . Despite the fact that many known neurotransmitters and neuropeptides are present in the ENS, only a handful are consistently used as markers to identify major cell types in the ENS. Some examples include nitric oxide synthase (NOS1) for IMNs and acetylcholine for EMNs as well as CALCA/B for IPANs 2 . While it is impressive how large swaths of ENS function are coordinated by these neurotransmitters and neuropeptides, it also raises the question of whether yet-to-be-determined factors confer cell-type specific functional properties 3 . These features might include secreted peptides and proteins, electrophysiological features mediated by ion channels and transporters, metabolic profiles, and specialized capabilities for cell-cell interaction or synapse formation. To accurately understand the identity of individual cell types, it is also crucial to consider the contextual information in which they operate, including their position in the gut. Histochemical and functional assays have identified differences along the length of the GI tract, as well as between species, highlighting the need for more systematic efforts towards comprehensive mapping and profiling of the ENS 1 ."

Multimodal Integration Studies

Multimodal integration studies have successfully aligned morphological, electrophysiological, neurochemical, and transcriptomic classifications of enteric neurons, validating traditional functional categories while revealing unprecedented molecular complexity. These integrative approaches demonstrate that Dogiel Type II morphology, AH-type electrophysiology, and specific gene expression patterns consistently identify the same neuronal populations across different classification modalities. (7 sources)

The integration of multiple classification modalities has been essential for establishing comprehensive frameworks for understanding enteric neuron diversity and has validated decades of functional research through molecular approaches (Fung et al., 2020). The evolution of enteric neuron classification began with Dogiel's morphological descriptions in the late 1800s, progressed through Hirst and colleagues' electrophysiological characterizations in 1974, and was expanded by Costa, Furness, and colleagues through extensive immunohistochemical and electron microscopy studies that grouped enteric neurons based on their distinct neurochemistry, projection patterns, and ultrastructure (Fung et al., 2020) (Costa et al., 1992).

Morpho-electro-chemical correlations have established robust multimodal alignments across classification systems. Intrinsic sensory neurons (IPANs) consistently display Dogiel Type II morphology, AH-type electrophysiological characteristics with prolonged afterhyperpolarization, and specific neurochemical profiles, demonstrating remarkable consistency across different analytical approaches (Fung et al., 2020). The afterhyperpolarization enables these neurons to set levels of excitability through somatic gating of activity within enteric networks, linking their electrophysiological properties directly to their functional roles (Fung et al., 2020).

Transcriptomic validation of classical classifications has provided molecular precision to traditional functional categories. Single-cell RNA sequencing studies have subdivided myenteric neurons into molecularly defined subpopulations that align closely with the ten different classes of myenteric neurons identified from integrated anatomical, neurochemical, functional, and pharmacological analysis (Fung et al., 2020). Recent work by Morarach and colleagues identified twelve enteric neuron classes within the myenteric plexus using single-cell sequencing, providing molecular evidence and histochemical markers for discrete classes of motor, sensory, and interneurons (Bianco et al., 2021) (Morarach et al., 2020).

Functional specialization across anatomical contexts demonstrates how multimodal approaches reveal location-specific properties. While IPANs can be identified using consistent morphological and electrophysiological criteria, single-cell sequencing data provides evidence for several subclasses with one clear functional division being whether they are situated in the submucosal or myenteric plexus, with only submucosal IPANs being sensitive to subtle mechanical stimulation of the mucosa (Fung et al., 2020).

Integration challenges and future directions highlight the ongoing need to connect deep transcriptional profiles with historical classification systems (Benthal et al., 2026). Despite decades of functional characterization identifying major cell types using established markers like nitric oxide synthase for inhibitory motor neurons and acetylcholine for excitatory motor neurons, the discovery of yet-to-be-determined factors that confer cell-type specific functional properties requires systematic integration efforts (Majd et al., 2024). These factors might include ion channels and transporters mediating electrophysiological features, metabolic profiles, and specialized capabilities for cell-cell interaction or synapse formation (Majd et al., 2024).

Comprehensive mapping efforts across species and gut regions emphasize the importance of contextual information in multimodal classification. Histochemical and functional assays have identified differences along the length of the gastrointestinal tract as well as between species, with 10-15 subtypes of myenteric neurons and 4-5 subtypes of submucosal neurons identified in the well-studied guinea pig ileum using combined morphological, neurochemical, electrophysiological, pharmacological, and tracing techniques (Majd et al., 2024) (Hao et al., 2009). Current computational integration efforts are reprocessing independent single-cell datasets to identify new cell types, shared marker genes across developmental stages, and achieve consensus on transcriptomic definitions of enteric neuronal subtypes (Benthal et al., 2026).

Evidence

(Fung et al., 2020)

Functional circuits and signal processing in the enteric nervous system

> *"Dogiel published illustrations of three morphological subtypes of neurons in the ENS, defined as Dogiel Types I, II, and III [14]. It was not until 1974 that Hirst and colleagues then reported on their electrophysiological properties and classified them into AH-and S-type neurons (Hirst et al., 1974). AH neurons are typified by their larger action potentials (APs) with an inflection on the falling phase and long afterhyperpolarizing potential (AHP; > 2 s) that follows. By comparison, S neurons are characterized by their brief APs that lack slow AHPs, and they display fast excitatory postsynaptic potentials (EPSPs)"*

> *"Costa, Furness, and colleagues conducted an extensive series of immunohistochemical and electron microscopy studies to group enteric neurons based on their distinct neurochemistry, projection patterns, and ultrastructure (Costa et al., 1996)(Costa et al., 1983)(Costa et al., 1981)(Costa et al., 1992)(Furness et al., 2004)(Furness et al., 1982)(Furness et al., 2004)[23][24][25][26][27][28][29]"*

> *"immunostaining for the expression of primary transmitters or their synthesizing enzymes, and cytoskeletal and calcium-binding proteins"*

> *"studies using various single-cell RNA-sequencing methods optimized for the ENS are now emerging, allowing for extensive gene expression profiling of functional molecular constituents of enteric neurons such as ion channels, transmitters and their synthesizing enzymes, as well as receptors [35,36]. This methodology based on statistical clustering algorithms subdivided myenteric neurons into nine molecularly defined subpopulations. By comparison, ten different classes of myenteric neurons were identified from integrated anatomical, neurochemical, functional, and pharmacological analysis (Furness et al., 2014)"*

> *"Intrinsic sensory neurons (or IPANs) are typically polymodal, possessing chemosensory and mechanosensory properties, and have Dogiel Type II morphology [40]. Furthermore, they are associated with AH-type characteristics where their AHP enables them to set a level of excitability through somatic gating of activity within the network [41]"*

> *"singlecell sequencing data provides evidence for several subclasses [35]36]. One clear division is whether they are situated in the submucosal or myenteric plexus. Only the former subset are sensitive to subtle mechanical stimulation of the mucosa"*

(Costa et al., 1992)

Projections and chemical coding of neurons with immunoreactivity for nitric oxide synthase in the guinea-pig small intestine.

> *"The distribution of nitric oxide synthase (NOS) immunoreactivity was investigated in the guinea-pig small intestine. There were many immunoreactive nerve cell bodies in the myenteric plexus but very few in submucous ganglia. NOS immunoreactivity was not found in non-neuronal cells except for rare mucosal endocrine cells. Abundant immunoreactive nerve fibres in both myenteric and submucous ganglia, and in the circular muscle, arose from myenteric nerve cells whose axons projected anally along the intestine. NOS immunoreactivity coexisted with VIP-immunoreactivity, but not with substance P immunoreactivity. We conclude that nitric oxide synthase is located in a sub-population of enteric neurons, amongst which are inhibitory motor neurons that supply the circular muscle layer."*

(Bianco et al., 2021)

Novel understanding on genetic mechanisms of enteric neuropathies leading to severe gut dysmotility

> *"Emerging research has provided novel insights on marker genes specific for the different enteric neuron classes (ENCs). Recently, Morarach et al. (Morarach et al., 2020) addressed this extremely innovative area and provided molecular evidence of twelve ENCs*

within the myenteric plexus of the mouse small intestine. The authors identified inter-communication features and histochemical markers for discrete classes of neurons, including motor, sensory, and interneurons together with genetic tools for class-specific targeting. (Morarach et al., 2020) A recent work from Ganz et al. (Ganz et al., 2019) showed that genes controlling epigenetic modifications are important to coordinate intestinal tract development, providing the first demonstration that these genes influence ENS development. In this line, another study in human and mouse healthy tissues used laser-capture microdissection coupled to single-nucleus RNA-sequencing, to map out enteric neuron subtypes in the duodenum, ileum, and colon, which were overall conserved between humans and mice based on orthologous gene expression. Nevertheless, some enteric neuron subtype-specific genes in mice were expressed in completely distinct morphologically defined neuron subtypes in humans, thus suggesting that these species-specific differences should be taken into account when translating findings from mouse to human ENS (May-Zhang et al., 2020) ."

(Morarach et al., 2020)

Diversification of molecularly defined myenteric neuron classes revealed by single cell RNA-sequencing

"Meticulous work over the last decades has identified a framework of enteric neurons with sensory (Intrinsic Primary Afferent Neurons, IPANs), motor or interneuron characteristics"

"we define a molecular taxonomy of 12 enteric neuron classes within the myenteric plexus of the mouse small intestine using single-cell RNA sequencing. We present cell–cell communication features and histochemical markers for motor neurons, sensory neurons and interneurons"

"some classes were defined by the selective expression of single genes with likely importance for their neuronal activity. These included ENC5 (somatostatin; Sst), ENC6 (neuromedin U; Nmu), ENC7 (cholecystokinin; Cck and urocortin 3; Ucn3), ENC10 (neuronal differentiation 6; Neurod6) and ENC12 (neurexophilin-2; Nxph2). Other classes were distinguished by combinatorial gene codes"

"ENC1-4 displayed a partly shared gene expression, including Tac1, but differed in their expression of Calb2, Ndufa4l2, Gda, Penk and Fut9"

"ENC8-9 jointly expressed Nos1 and c1ql1, but Npy and Cox8b were significantly lower in ENC9. ENC11 highly expressed Npy, but lacked Nos1"

"ENC1-4 were assigned to excitatory motor neurons (Tac1/Calb2), while ENC8 and 9 matched inhibitory motor neurons (Nos1/Gal/Vip/Npy). ENC6, 7 and 12 were mapped to IPANs and expressed different combinations of previously defined markers (Calca/Calcb/Nfem/Calb1/Calb2). These plausible IPANs were clearly separable by their selective expression of Nmu, Ucn-3/Cck or Nxph2."

(Benthal et al., 2026)

Building consensus: construction of a juvenile and adult scRNA-seq meta-atlas for dataset comparisons and harmonizing transcriptomic definitions of enteric neuron subtypes

"Previously, distinct enteric neuron types were cataloged by dye-filling techniques, immunohistochemistry, retrograde labeling, and electrophysiology. Recent technical advances in single cell RNA-sequencing (scRNA-seq) have enabled transcriptional profiling of hundreds to millions of individual cells from the intestine. These data allow cell types to be resolved and compared using their transcriptional profiles ("clusters") rather than relying on antibody labeling"

"Here we briefly systematically review each Mus musculus single cell or single nucleus RNA-sequencing (snRNA-seq) ENS dataset. We then reprocess and computationally integrate these select independent scRNA-seq enteric neuron datasets from the myenteric plexus with the aim to identify new cell types, shared marker genes across juvenile to adult ages, dataset differences, and achieve some consensus on transcriptomic definitions of enteric neuronal subtypes"

"Future studies face the challenge of connecting these deep transcriptional profiles for enteric neurons with historical classification systems."

(Majd et al., 2024)

A call for a unified and multimodal definition of cellular identity in the enteric nervous system

"For decades, enteric neuron subtype identity has largely been defined based on morphological, immunohistochemical and functional features of neurons within the gut, including neuronal projection localization, neurochemistry, and electrophysiological properties. Investigations conducted on tissues isolated from model organisms, such as the guinea pig, mouse and rat 1 have significantly progressed our understanding of the neurochemical and functional complexity of enteric neuron subtypes and have identified functional units within the ENS circuitry. These include intrinsic primary afferent neurons (IPANs or sensory neurons, SNs), ascending and descending interneurons (IN), excitatory and inhibitory motor neurons (EMNs and IMNs), and secretomotor and/or vasodilator neurons (SVNs) 1 . Despite the fact that many known neurotransmitters and neuropeptides are present in the ENS, only a handful are consistently used as markers to identify major cell types in the ENS. Some examples include nitric oxide synthase (NOS1) for IMNs and acetylcholine for EMNs as well as CALCA/B for IPANs 2 . While it is impressive how large swaths of ENS

function are coordinated by these neurotransmitters and neuropeptides, it also raises the question of whether yet-to-be-determined factors confer cell-type specific functional properties 3 . These features might include secreted peptides and proteins, electrophysiological features mediated by ion channels and transporters, metabolic profiles, and specialized capabilities for cell-cell interaction or synapse formation. To accurately understand the identity of individual cell types, it is also crucial to consider the contextual information in which they operate, including their position in the gut. Histochemical and functional assays have identified differences along the length of the GI tract, as well as between species, highlighting the need for more systematic efforts towards comprehensive mapping and profiling of the ENS 1 ."

(Hao et al., 2009)

Development of enteric neuron diversity

"Initially, enteric neuron subtypes were identified by Dogiel based on morphological characteristics alone (see [1]). However, elegant studies performed in the past few decades utilizing a variety of immunohistochemical, electrophysiological, pharmacological and tracing techniques have identified different subtypes of enteric neurons based on a combination of morphology, neurotransmitters, electrophysiology, target tissue and direction (up, down or circumferentially) and length of axon projection.

Enteric neurons have been most thoroughly studied in the guinea-pig ileum, where 10-15 subtypes of myenteric neurons and 4-5 subtypes of submucosal neurons have been identified; these include intrinsic primary afferent (sensory) neurons, ascending and descending interneurons, excitatory and inhibitory motor neurons to both the longitudinal and circular muscle layers, intestinofugal, secretomotor and vasodilator neurons [5][6][7]"

"In mammals and birds, most enteric neurons are clustered in myenteric ganglia (Fig. 1A), which are located between the circular and longitudinal muscle layers, and in submucosal ganglia. In the intestine of smaller mammals, myenteric neurons are primarily involved in the regulation of gut motility and submucosal neurons are mostly involved in the regulation of secretion and vascular tone; in larger mammals, some submucosal neurons are also directly involved in motility reflexes [4]."